

Phase 1 - Excel

Excel Day 1

Spreadsheet Beginnings

Spreadsheet beginnings:

- Replace paper ledger books for accounting
- Tabular data
- Early 70's LANPAR (LANguage for Programming Arrays at Random)
- 1979 - Visicalc for PCs
- 1985 - Excel 1.0 for Mac

Why Excel for DSC 270?

Why Excel and not Google Sheets?

- Industry standard
- Data can be local
- Has more advanced features
- Can handle more data
- More keyboard shortcuts

See <https://zapier.com/blog/google-sheets-vs-excel/> for detailed comparison

Warnings!!!

Desktop version vs. web-based version

More than 1 way to do things - can't show all

If You “Know” Excel Already, Bear With Me

Most students don't know Excel

- Every intro class I've taught $\geq 90\%$ of students can't do a simple sum
- Be patient, only 4 days, you'll learn something!
- 4 days just scratches the surface, be prepared to learn independently on your own in future (Help menu, STFW, learn to write a good search string and refine bad ones)

I've used it for > 20 years, so stop me and ask if I go too fast!

Excel & Windows 11

Campus changed to Windows 11

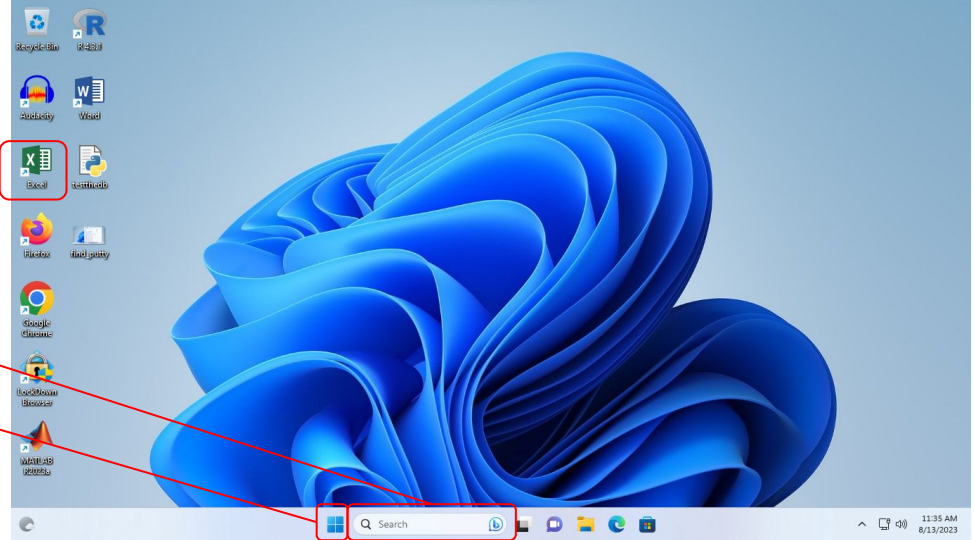
Shortcut on desktop

Search bar

Windows menu

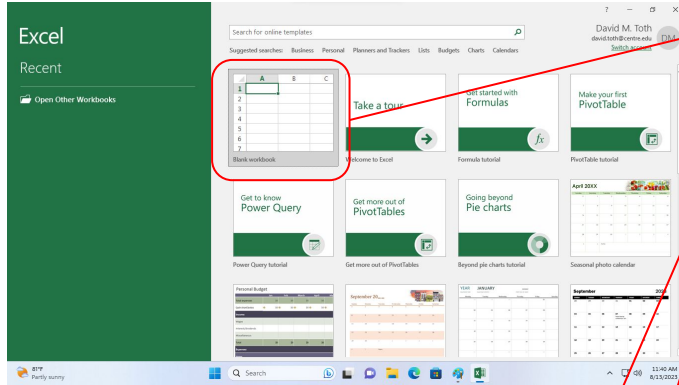
Double click an .xlsx file

File associations

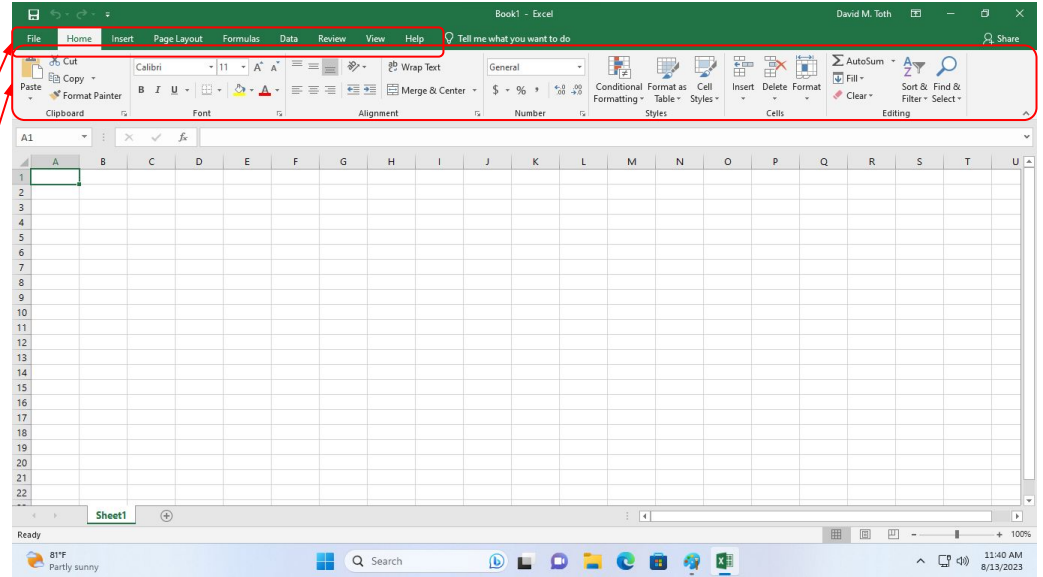


In class, use provided computers

When Excel Starts



Menus
Help
Tool ribbon



Keyboard Shortcuts

Know these!

My rant: don't copy, paste, & delete instead of cut & paste!

Copy: control-c

Cut: control-x

Paste: control-v

Undo: control-z

Redo: control-y

Learning Styles

List all tools & how to use them and then get told to build this

OR

Get a task and learn the tools on the fly
“just-in-time”

Formulas
Absolute and relative cell references
Freeze rows & cols
Weighted average example
Filling/copying formulas
Basic statistics
Insert/delete row, col
Conditional formatting
Sorting data
Make graphs

Build a gradebook for UC Berkeley’s equivalent to this class (hundreds of students)

STOP - ACTUALLY BUILD THE GRADEBOOK!!!

First I demo how to build it.

Next students do activity to cement the things they've learned.



Formulas

=(blah blah blah)

Do simple sum example

Cell References

(rowcol) like A5

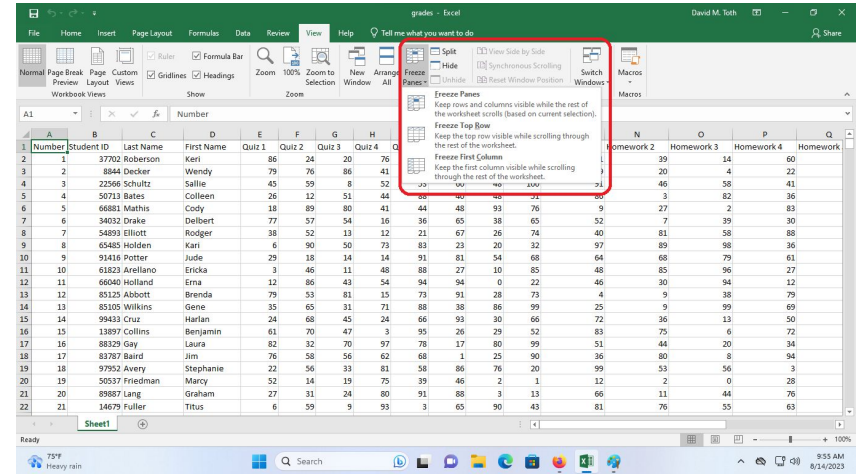
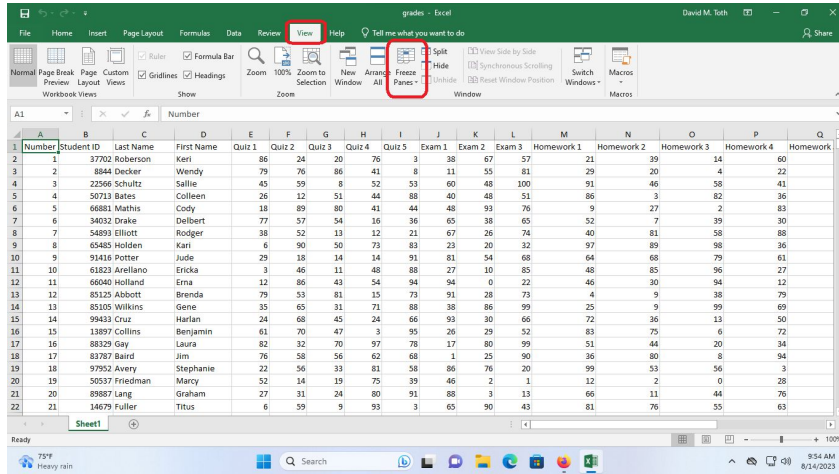
Absolute - has \$

Relative - no \$

Do quick example!

Freezing Rows & Columns

Can freeze top row or first column OR



Select the cell down and right of what you want to freeze

Freezing Rows & Columns

Select the cell down and right of what you want to freeze and select Freeze Panes

The screenshot shows the Microsoft Excel interface with the 'View' tab selected in the ribbon. The 'Freeze Panes' option is highlighted in the ribbon, and a red box highlights the 'Freeze Panes' option and the 'Freeze Top Row' and 'Freeze First Column' options. The spreadsheet data is visible in the background.

Number	Student ID	Last Name	First Name	Quiz 1	Quiz 2	Quiz 3	Quiz 4	Quiz 5	Quiz 6	Quiz 7	Quiz 8	Quiz 9	Quiz 10	Quiz 11	Quiz 12	Quiz 13	Quiz 14	Quiz 15	Quiz 16	Quiz 17	Quiz 18	Quiz 19	Quiz 20	Quiz 21	Quiz 22	Quiz 23	Quiz 24	Quiz 25	Quiz 26	Quiz 27	Quiz 28	Quiz 29	Quiz 30	Quiz 31	Quiz 32	Quiz 33	Quiz 34	Quiz 35	Quiz 36	Quiz 37	Quiz 38	Quiz 39	Quiz 40	Quiz 41	Quiz 42	Quiz 43	Quiz 44	Quiz 45	Quiz 46	Quiz 47	Quiz 48	Quiz 49	Quiz 50	Quiz 51	Quiz 52	Quiz 53	Quiz 54	Quiz 55	Quiz 56	Quiz 57	Quiz 58	Quiz 59	Quiz 60	Quiz 61	Quiz 62	Quiz 63	Quiz 64	Quiz 65	Quiz 66	Quiz 67	Quiz 68	Quiz 69	Quiz 70	Quiz 71	Quiz 72	Quiz 73	Quiz 74	Quiz 75	Quiz 76	Quiz 77	Quiz 78	Quiz 79	Quiz 80	Quiz 81	Quiz 82	Quiz 83	Quiz 84	Quiz 85	Quiz 86	Quiz 87	Quiz 88	Quiz 89	Quiz 90	Quiz 91	Quiz 92	Quiz 93	Quiz 94	Quiz 95	Quiz 96	Quiz 97	Quiz 98	Quiz 99	Quiz 100
1	37702	Roberson	Keri	86	2	20	7	4	88	40	48	51	86	39	14	60	20	4	22	46	58	41	3	82	36	9	27	2	83	52	7	39	30	40	81	58	88	97	89	98	36	64	68	79	61	48	85	96	27	46	30	94	12	4	9	38	79	25	9	99	69	72	36	13	50	83	75	6	72	51	44	20	34	94	8	3	12	2	0	28	66	11	44	76	81	76	55	63															

Unfreezing Panes

The screenshot shows the Microsoft Excel interface. The 'View' ribbon is active, displaying options like 'Ruler', 'Formula Bar', 'Gridlines', 'Headings', 'Zoom', 'New Window', and 'Arrange All'. A red box highlights the 'Unfreeze Panes' context menu, which is open over the worksheet. The menu options are: 'Unfreeze Panes' (Unlock all rows and columns to scroll through the entire worksheet.), 'Freeze Top Row' (Keep the top row visible while scrolling through the rest of the worksheet.), and 'Freeze First Column' (Keep the first column visible while scrolling through the rest of the worksheet.). The worksheet contains a table with columns: Student ID, Last Name, First Name, Quiz 1, Quiz 2, Quiz 3, Quiz 4, Homework 2, Homework 3, Homework 4, and Homework. The data is organized into rows, with the first row being the header and subsequent rows containing student information and quiz scores.

Weighted Average

5 quizzes (cols E-I)

3 exams (cols J-L)

10 homeworks (cols M-V)

Assume course grade is 20% homework average, 30% quiz average, and 50% exam average

Make cols W, X, and Y the averages of the quizzes, exams, and homeworks

Make col Z Course Grade

New Columns

The screenshot shows an Excel spreadsheet titled "grades - Excel" with a green ribbon. The "Home" tab is active, showing the ribbon with options like Clipboard, Font, Alignment, Number, Styles, Cells, and Editing. The spreadsheet has columns A through Z. Columns W, X, Y, and Z are highlighted with a red box. The data in these columns is as follows:

	W	X	Y	Z
1	Quiz Avg	Exam Avg	Homework Avg	Course Grade
2	51			
3				
4				
5				
6				
7				
8				
9				
10				
11				
12				
13				
14				
15				
16				
17				
18				
19				
20				
21				
22				

The status bar at the bottom shows "Ready", "73°F Heavy rain", and the system clock "10:15 AM 8/14/2023".

Putting In The Averages

In cell W2, type **= AVERAGE(E2:I2)** and hit the Enter key.

In cell X2, type **= AVERAGE(J2:L2)** and hit the Enter key.

In cell Y2, type **= AVERAGE(** and then scroll to the left using the scrollbar, left-click and hold in cell M2, drag to the right until you have M2 to V2 selected, then release the mouse button and type **)** and hit the Enter key.

In cell Z2, type **= (.2*W2) + (.3*X2) + (.5*Y2)** and hit the Enter key.

Filling/Copying Formulas

- select W2 to Z2, grab the green square at the bottom of Z2, drag it down, and release
- or double-click the green square
- or copy and paste
- or (next slide)

grades - Excel

David M. Toth

File Home Insert Page Layout Formulas Data Review View Help Tell me what you want to do

Clipboard Font Alignment Number Styles Cells Editing

W2 =AVERAGE(E2:I2)

	A	B	C	D	Q	R	S	T	U	V	W	X	Y	Z	
	Number	Student ID	Last Name	First Name	Homework 5	Homework 6	Homework 7	Homework 8	Homework 9	Homework 10	Quiz Avg	Exam Avg	Homework Avg	Courses	Grade
2	1	26918	Roberson	Keri	24	80	20	53	69	23	66.6	67.66667	40.2		53.72
3	2	65472	Decker	Wendy	24	42	41	78	62	60					
4	3	14599	Schultz	Sallie	39	79	57	67	58	13					
5	4	70183	Bates	Colleen	85	18	74	47	3	53					
6	5	69456	Mathis	Cody	51	73	37	48	46	90					
7	6	16923	Drake	Delbert	28	75	87	7	50	12					
8	7	32741	Elliott	Rodger	13	29	86	97	44	76					
9	8	71786	Holden	Kari	44	100	64	48	96	52					
10	9	19240	Potter	Jude	77	69	47	13	26	88					
11	10	22179	Arellano	Ericka	13	28	60	69	25	17					
12	11	33934	Holland	Erna	69	57	66	68	28	55					
13	12	8345	Abbott	Brenda	81	11	45	54	4	42					
14	13	47524	Wilkins	Gene	6	30	74	28	14	64					
15	14	75701	Cruz	Harlan	89	5	13	86	56	32					
16	15	89611	Collins	Benjamin	59	65	30	65	83	74					
17	16	70771	Gay	Laura	76	29	63	1	45	88					
18	17	28391	Baird	Jim	61	41	19	7	24	40					
19	18	35730	Avery	Stephanie	53	84	39	92	33	77					
20	19	62232	Friedman	Marcy	57	23	85	23	29	37					
21	20	83128	Lang	Graham	91	16	15	80	92	67					
22	21	30072	Fuller	Titus	78	85	13	24	43	56					

Sheet1

Ready Average: 57.0466667 Count: 4 Sum: 228.1866667

73°F Heavy rain 10:27 AM 8/14/2023

Filling/Copying Formulas

Fill formulas into adjacent cells

You can use the **Fill** command to fill a formula into an adjacent range of cells. Simply do the following:

1. Select the cell with the formula and the adjacent cells you want to fill.
2. Click **Home** > **Fill**, and choose either **Down**, **Right**, **Up**, or **Left**.

Keyboard shortcut: You can also press Ctrl+D to fill the formula down in a column, or Ctrl+R to fill the formula to the right in a row.

Basic Statistics

In cell E503, type **= MEDIAN(E2:E501)** and hit the Enter key.

In cell E504, type **= MODE(E2:E501)** and hit the Enter key.

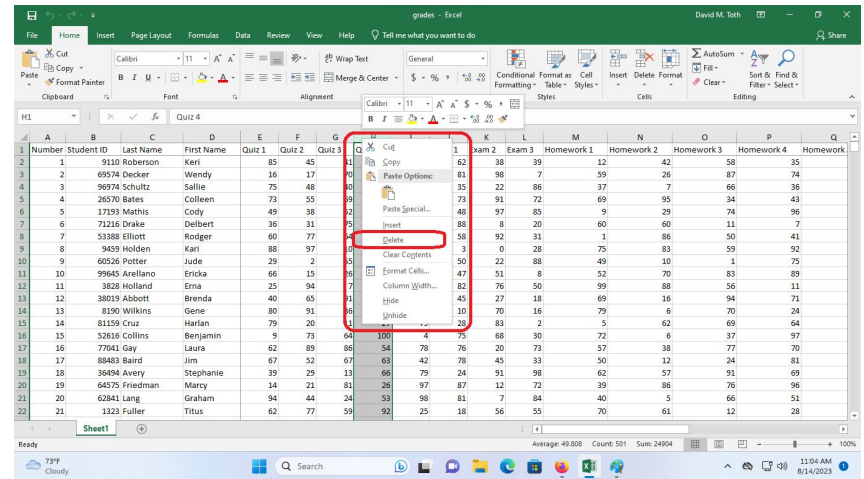
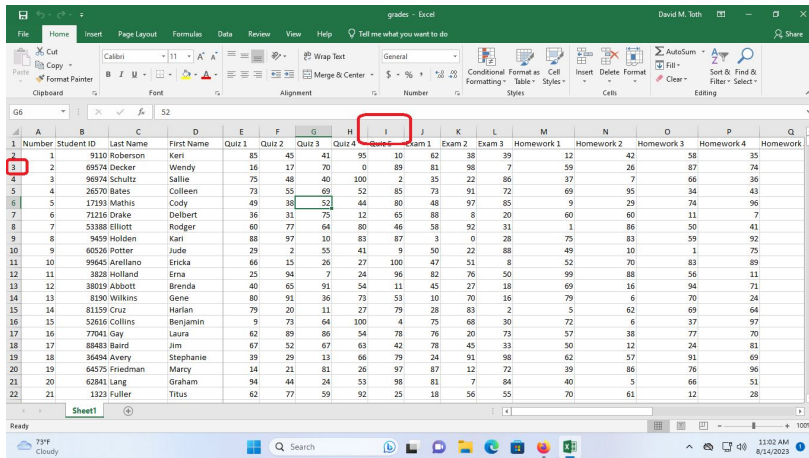
In cell E505, type **= MAX(E2:E501)** and hit the Enter key.

In cell E506, type **= MIN(E2:E501)** and hit the Enter key.

Can do lots of normal statistic functions like standard deviation, etc.

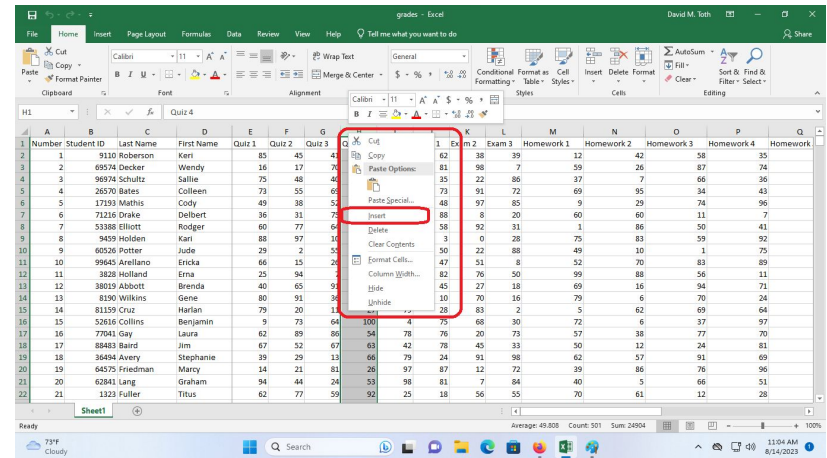
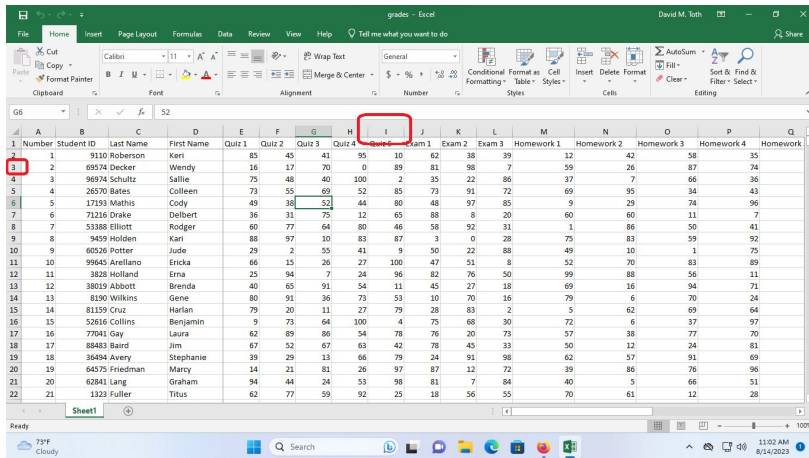
Deleting A Row Or Column

Right click on a row's header or a column's header and select delete from the pop-up menu.



Inserting A Row Or Column

Right click on a row's header or a column's header and select insert from the pop-up menu.



Conditional Formatting

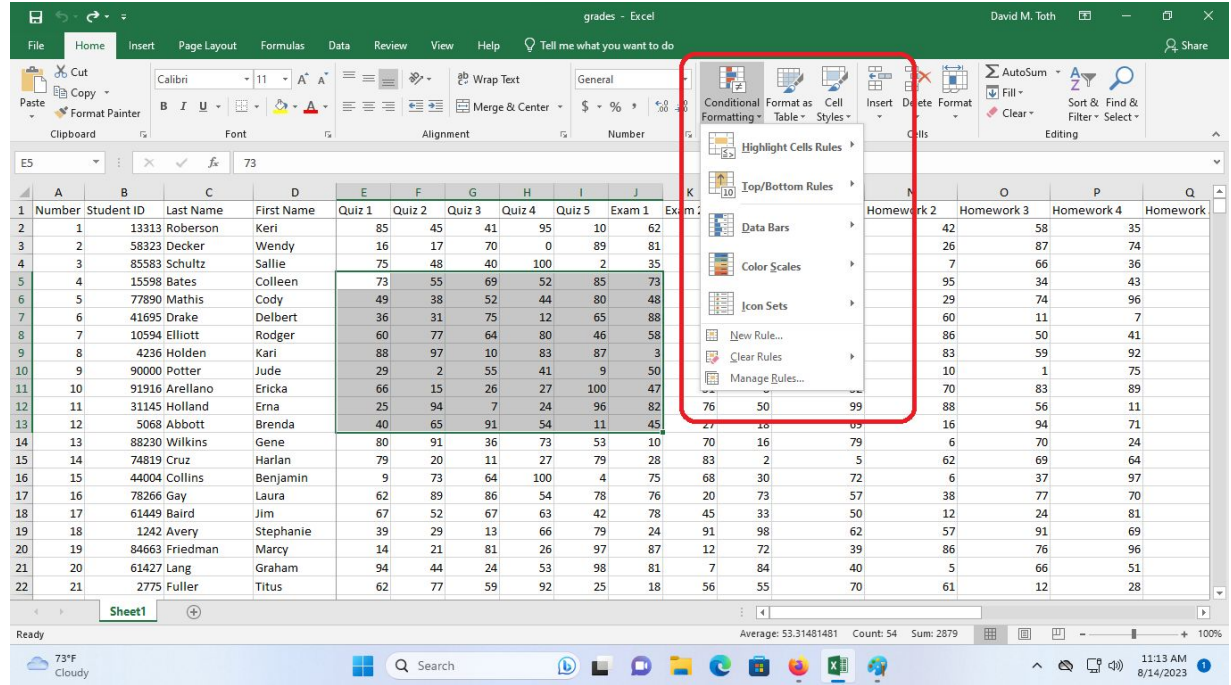
The screenshot shows the Microsoft Excel interface with the 'Home' tab selected on the ribbon. The 'Conditional Formatting' button is highlighted with a red box. The spreadsheet displays a table of student data with columns for Name, Quiz scores, Exam scores, and Homework scores. The cell K13 (row 13, column K) is highlighted with a green border.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q
	Number	Student ID	Last Name	First Name	Quiz 1	Quiz 2	Quiz 3	Quiz 4	Quiz 5	Exam 1	Exam 2	Exam 3	Homework 1	Homework 2	Homework 3	Homework 4	Homework 5
1	1	13313	Roberson	Keri	85	45	41	95	10	62	38	39	12	42	58	35	
2	2	58323	Decker	Wendy	16	17	70	0	89	81	98	7	59	26	87	74	
3	3	85583	Schultz	Sallie	75	48	40	100	2	35	22	86	37	7	66	36	
4	4	15598	Bates	Colleen	73	55	69	52	85	73	91	72	69	95	34	43	
5	5	77890	Mathis	Cody	49	38	52	44	80	48	97	85	9	29	74	96	
6	6	41695	Drake	Delbert	36	31	75	12	65	88	8	20	60	60	11	7	
7	7	10594	Elliott	Rodger	60	77	64	80	46	58	92	31	1	86	50	41	
8	8	4236	Holden	Karl	88	97	10	83	87	3	0	28	75	83	59	92	
9	9	90000	Potter	Jude	29	2	55	41	9	50	22	88	49	10	1	75	
10	10	91916	Arellano	Ericka	66	15	26	27	100	47	51	8	52	70	83	89	
11	11	31145	Holland	Erna	25	94	7	24	96	82	76	50	99	88	56	11	
12	12	5068	Abbott	Brenda	40	65	91	54	11	45	27	18	69	16	94	71	
13	13	88230	Wilkins	Gene	80	91	36	73	53	10	70	16	79	6	70	24	
14	14	74819	Cruz	Harlan	79	20	11	27	79	28	83	2	5	62	69	64	
15	15	44004	Collins	Benjamin	9	73	64	100	4	75	68	30	72	6	37	97	
16	16	78266	Gay	Laura	62	89	86	54	78	76	20	73	57	38	77	70	
17	17	61449	Baird	Jim	67	52	67	63	42	78	45	33	50	12	24	81	
18	18	1242	Avery	Stephanie	39	29	13	66	79	24	91	98	62	57	91	69	
19	19	84663	Friedman	Marcy	14	21	81	26	97	87	12	72	39	86	76	96	
20	20	61427	Lang	Graham	94	44	24	53	98	81	7	84	40	5	66	51	
21	21	2775	Fuller	Titus	62	77	59	92	25	18	56	55	70	61	12	28	

Conditional Formatting

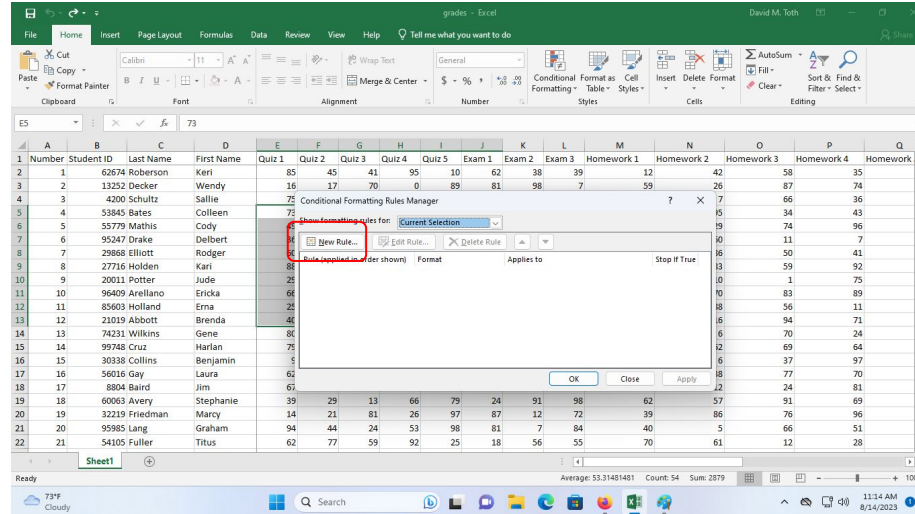
Select the data to
format

Click the conditional
formatting button



Conditional Formatting

Click new rule



Conditional Formatting

Get this screen

The screenshot shows the Microsoft Excel interface with a spreadsheet titled 'grades - Excel'. The spreadsheet has columns for student information and quiz scores. A 'New Formatting Rule' dialog box is open, showing the 'Format all cells based on their values' rule type. The rule is configured to format cells with values between 40 and 62 using a 2-color scale from orange to yellow.

Spreadsheet Data:

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	
1	Number	Student ID	Last Name	First Name	Quiz 1	Quiz 2	New Formatting Rule								Homework 2	Homework 3	Homework 4	Homework 5
2	1	39571	Roberson	Keri	85									42	58	35		
3	2	99160	Decker	Wendy	16									26	87	74		
4	3	50787	Schultz	Sallie	75									7	66	36		
5	4	73038	Bates	Colleen	73									95	34	43		
6	5	19995	Mathis	Cody	49									29	74	96		
7	6	54276	Drake	Delbert	36									60	11	7		
8	7	41219	Elliott	Rodger	60									86	50	41		
9	8	9875	Holden	Kari	88									83	59	92		
10	9	24342	Potter	Jude	29									10	1	75		
11	10	34849	Arellano	Ericka	66									70	83	89		
12	11	41881	Holland	Erna	25									88	56	11		
13	12	93808	Abbott	Brenda	40									16	94	71		
14	13	62261	Wilkins	Gene	80									6	70	24		
15	14	48804	Cruz	Harlan	79									62	69	64		
16	15	82038	Collins	Benjamin	9									6	37	97		
17	16	46430	Gay	Laura	62									38	77	70		
18	17	48563	Baird	Jim	67									12	24	81		
19	18	7959	Avery	Stephanie	39									57	91	69		
20	19	7592	Friedman	Marcy	14									86	76	96		
21	20	79994	Lang	Graham	94		44	24	53	98	81	7	84	40	5	66	51	
22	21	35989	Fuller	Titus	62		77	59	92	25	18	56	55	70	61	12	28	

Conditional Formatting Rule Details:

- Select a Rule Type:** Format all cells based on their values
- Edit the Rule Description:** Format all cells based on their values
- Format Style:** 2-Color Scale
- Type:** Lowest Value (Minimum), Highest Value (Maximum)
- Value:** (Lowest value) 40, (Highest value) 62
- Color:** Orange (Minimum), Yellow (Maximum)
- Preview:** A gradient bar from orange to yellow.

Conditional Formatting

Choose values and
click Format...
button

The screenshot displays the Microsoft Excel interface with a spreadsheet titled 'grades - Excel'. The 'Home' tab is active, and the 'Conditional Formatting' button in the 'Styles' group is visible. A 'New Formatting Rule' dialog box is open, showing the following configuration:

- Select a Rule Type:** Format only cells that contain
- Edit the Rule Description:** Format only cells with:
 - Cell Value
 - less than or equal to
 - 60
- Preview:** No Format Set
- Buttons:** OK, Cancel, and a highlighted 'Format...' button.

The background spreadsheet contains student data and quiz scores. The columns are labeled: Number, Student ID, Last Name, First Name, Quiz 1, Quiz 2, and Homework 2 through Homework 5. The data rows show scores for 22 students. The 'Format...' button in the dialog is highlighted with a red box, indicating the next step in the process.

Conditional Formatting

Get this

The screenshot shows the Microsoft Excel interface with a spreadsheet titled 'grades - Excel'. The spreadsheet contains student data with columns for Number, Student ID, Last Name, First Name, Quiz 1, and Quiz 2. The 'Format Cells' dialog box is open, showing the 'Fill' tab with a red background color selected. The status bar at the bottom indicates 'Ready' and 'Average: 52.80952381 Count: 105 Sum: 5545'.

Number	Student ID	Last Name	First Name	Quiz 1	Quiz 2
1	39571	Roberson	Keri	85	
2	99160	Decker	Wendy	16	
3	50787	Schultz	Sallie	75	
4	73038	Bates	Colleen	73	
5	19995	Mathis	Cody	49	
6	54276	Drake	Delbert	36	
7	41219	Elliott	Rodger	60	
8	9875	Holden	Kari	88	
9	24342	Potter	Jude	29	
10	34849	Arellano	Ericka	66	
11	41881	Holland	Erna	25	
12	93808	Abbott	Brenda	40	
13	62261	Wilkins	Gene	80	
14	48804	Cruz	Harlan	79	
15	82038	Collins	Benjamin	9	
16	46430	Gay	Laura	62	
17	48563	Baird	Jim	67	
18	7959	Avery	Stephanie	39	
19	7592	Friedman	Marcy	14	
20	79994	Lang	Graham	94	
21	35989	Fuller	Titus	62	

Sheet1

Ready 74°F Mostly sunny

Average: 52.80952381 Count: 105 Sum: 5545

11:47 AM 8/15/2023

Conditional Formatting

Voila!

grades - Excel

David M. Toth

File Home Insert Page Layout Formulas Data Review View Help Tell me what you want to do

Clipboard Font Paragraph Styles Cells Editing

Conditional Formatting Format as Table Cell Styles Insert Delete Format AutoSum Fill Clear Sort & Find & Filter Select

E2 85

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q
1	Number	Student ID	Last Name	First Name	Quiz 1	Quiz 2	Quiz 3	Quiz 4	Quiz 5	Exam 1	Exam 2	Exam 3	Homework 1	Homework 2	Homework 3	Homework 4	Homework
2	1	39571	Roberson	Keri	85	45	41	95	16	62	38	39	12	42	58	35	
3	2	99160	Decker	Wendy	16	17	70	0	89	81	98	7	59	26	87	74	
4	3	50787	Schultz	Sallie	75	48	48	100	2	33	20	86	37	7	66	36	
5	4	73038	Bates	Colleen	73	55	69	52	85	73	91	72	69	95	34	43	
6	5	19995	Mathis	Cody	49	38	52	44	80	48	97	85	9	29	74	96	
7	6	54276	Drake	Delbert	38	31	75	12	65	88	3	20	60	60	11	7	
8	7	41219	Elliott	Rodger	60	77	64	80	46	58	92	31	1	86	50	41	
9	8	9875	Holden	Kari	88	97	18	83	87	3	0	28	75	83	59	92	
10	9	24342	Potter	Jude	29	2	55	41	9	50	22	88	49	10	1	75	
11	10	34849	Arellano	Ericka	66	15	26	27	100	47	59	8	52	70	83	89	
12	11	41881	Holland	Erna	25	94	7	24	96	82	76	50	99	88	56	11	
13	12	93808	Abbott	Brenda	40	65	91	54	11	45	29	18	69	16	94	71	
14	13	62261	Wilkins	Gene	80	91	36	73	53	10	70	16	79	6	70	24	
15	14	48804	Cruz	Harlan	79	20	13	27	79	28	83	2	5	62	69	64	
16	15	82038	Collins	Benjamin	5	73	64	100	4	75	68	30	72	6	37	97	
17	16	46430	Gay	Laura	62	89	86	54	78	76	20	73	57	38	77	70	
18	17	48563	Baird	Jim	67	52	67	63	42	78	45	33	50	12	24	81	
19	18	7959	Avery	Stephanie	39	29	13	66	79	24	91	98	62	57	91	69	
20	19	7592	Friedman	Marcy	14	21	81	26	97	87	12	72	39	86	76	96	
21	20	79994	Lang	Graham	94	44	24	53	98	81	7	84	40	5	66	51	
22	21	35989	Fuller	Titus	62	77	59	92	25	18	56	55	70	61	12	28	

Sheet1

Ready Average: 52.80952381 Count: 105 Sum: 5545 74°F Mostly sunny 11:47 AM 8/15/2023

Conditional Formatting

Can add more rules, but make sure to order them correctly or you won't get what you think!

The screenshot displays the Microsoft Excel interface with a spreadsheet titled 'grades - Excel'. The spreadsheet contains student data and test scores. A 'Conditional Formatting Rules Manager' dialog box is open, showing two rules:

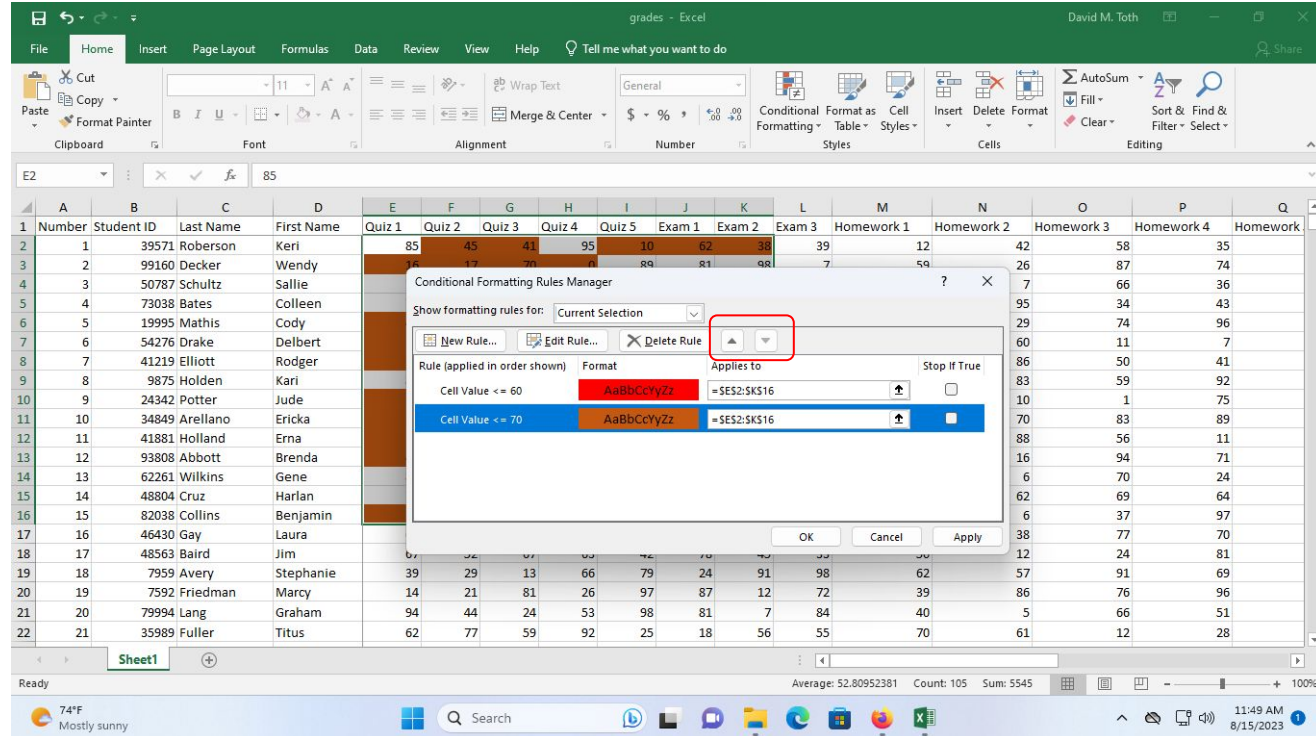
- Rule 1: Cell Value <= 70, Format: AaBbCcYyZz, Applies to: =\$E\$2:\$K\$16, Stop If True: [X]
- Rule 2: Cell Value <= 60, Format: AaBbCcYyZz, Applies to: =\$E\$2:\$K\$16, Stop If True: []

The background spreadsheet shows columns for Student ID, Last Name, First Name, and various test scores (Quiz 1, Quiz 2, Quiz 3, Quiz 4, Quiz 5, Exam 1, Exam 2, Exam 3, Homework 1, Homework 2, Homework 3, Homework 4). The scores are color-coded based on the conditional formatting rules.

At the bottom of the screen, the Windows taskbar shows the system clock as 11:48 AM on 8/15/2023.

Conditional Formatting

Order the rules
using these buttons



Conditional Formatting Rules Manager

Show formatting rules for: Current Selection

New Rule... Edit Rule... Delete Rule

Rule (applied in order shown) Format Applies to Stop If True

Cell Value <= 60 AaBbCcYyZz = \$E\$2:\$K\$16

Cell Value <= 70 AaBbCcYyZz = \$E\$2:\$K\$16

OK Cancel Apply

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q
1	Number	Student ID	Last Name	First Name	Quiz 1	Quiz 2	Quiz 3	Quiz 4	Quiz 5	Exam 1	Exam 2	Exam 3	Homework 1	Homework 2	Homework 3	Homework 4	Homework 5
2	1	39571	Roberson	Keri	85	45	41	95	10	62	38	39	12	42	58	35	
3	2	99160	Decker	Wendy	16	17	20	0	89	81	98	7	59	26	87	74	
4	3	50787	Schultz	Sallie										7	66	36	
5	4	73038	Bates	Colleen											95	34	43
6	5	19995	Mathis	Cody											29	74	96
7	6	54276	Drake	Delbert											60	11	7
8	7	41219	Elliott	Rodger											86	50	41
9	8	9875	Holden	Kari											83	59	92
10	9	24342	Potter	Jude											10	1	75
11	10	34849	Arellano	Ericka											70	83	89
12	11	41881	Holland	Erna											88	56	11
13	12	93808	Abbott	Brenda											16	94	71
14	13	62261	Wilkins	Gene											6	70	24
15	14	48804	Cruz	Harlan											62	69	64
16	15	82038	Collins	Benjamin											6	37	97
17	16	46430	Gay	Laura											38	77	70
18	17	48563	Baird	Jim											12	24	81
19	18	7959	Avery	Stephanie		29	13	66	79	24	91	98	62	57	91	69	
20	19	7592	Friedman	Marcy		21	81	26	97	87	12	72	39	86	76	96	
21	20	79994	Lang	Graham		44	24	53	98	81	7	84	40	5	66	51	
22	21	35989	Fuller	Titus		77	59	92	25	18	56	55	70	61	12	28	

Ready 74°F Mostly sunny Average: 52.80952381 Count: 105 Sum: 5545 11:49 AM 8/15/2023

Conditional Formatting

This is what
we wanted

grades - Excel

David M. Toth

File Home Insert Page Layout Formulas Data Review View Help Tell me what you want to do

Clipboard Font Alignment Number Styles

Conditional Formatting Format as Table Cell Styles Insert Delete Format AutoSum Fill Sort & Find & Filter

E2 85

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q
1	Number	Student ID	Last Name	First Name	Quiz 1	Quiz 2	Quiz 3	Quiz 4	Quiz 5	Exam 1	Exam 2	Exam 3	Homework 1	Homework 2	Homework 3	Homework 4	Homework 5
2	1	39571	Roberson	Keri	85	45	41	95	10	62	11	39	12	42	58	35	
3	2	99160	Decker	Wendy	18	17	70	6	89	81	98	7	59	26	87	74	
4	3	50787	Schultz	Sallie	75	48	40	100	2	35	22	86	37	7	66	36	
5	4	73038	Bates	Colleen	73	55	69	52	85	73	91	72	69	95	34	43	
6	5	19995	Mathis	Cody	49	38	52	44	80	42	97	85	9	29	74	96	
7	6	54276	Drake	Delbert	36	84	75	12	65	88	8	20	60	60	11	7	
8	7	41219	Elliott	Rodger	60	77	64	80	46	58	92	31	1	86	50	41	
9	8	9875	Holden	Kari	88	97	10	83	87	3	8	28	75	83	59	92	
10	9	24342	Potter	Jude	29	2	55	41	9	50	24	88	49	10	1	75	
11	10	34849	Arellano	Ericka	66	15	26	27	100	47	54	8	52	70	83	89	
12	11	41881	Holland	Erna	25	94	7	24	96	82	76	50	99	88	56	11	
13	12	93808	Abbott	Brenda	40	65	91	54	11	45	27	18	69	16	94	71	
14	13	62261	Wilkins	Gene	80	91	38	73	53	10	70	16	79	6	70	24	
15	14	48804	Cruz	Harlan	79	20	11	27	79	28	83	2	5	62	69	64	
16	15	82038	Collins	Benjamin	9	73	64	100	4	75	68	30	72	6	37	97	
17	16	46430	Gay	Laura	62	89	86	54	78	76	20	73	57	38	77	70	
18	17	48563	Baird	Jim	67	52	67	63	42	78	45	33	50	12	24	81	
19	18	7959	Avery	Stephanie	39	29	13	66	79	24	91	98	62	57	91	69	
20	19	7592	Friedman	Marcy	14	21	81	26	97	87	12	72	39	86	76	96	
21	20	79994	Lang	Graham	94	44	24	53	98	81	7	84	40	5	66	51	
22	21	35989	Fuller	Titus	62	77	59	92	25	18	56	55	70	61	12	28	

Sheet1

Ready Average: 52.80952381 Count: 105 Sum: 5545 74°F Mostly sunny 11:49 AM 8/15/2023

Sorting Data

Select the data you want to sort (careful to get everything)

The screenshot shows the Microsoft Excel interface with the 'Data' tab selected in the ribbon. The 'Sort & Filter' group is visible, with the 'Sort' button highlighted by a red box. The data table below contains student scores across various quizzes and homework assignments.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q
1	Number	Student ID	Last Name	First Name	Quiz 1	Quiz 2	Quiz 3	Quiz 4	Quiz 5	Exam 1	Exam 2	Exam 3	Homework 1	Homework 2	Homework 3	Homework 4	Homework
466	465	54980	Contreras	Malinda	66	5	60	8	5	25	77	72	99	62	26	76	
467	466	70180	Carlson	Columbus	33	40	25	61	0	82	28	29	6	74	45	97	
468	467	9224	Coleman	Cedric	21	0	20	66	50	34	37	0	20	45	83	59	
469	468	51552	Richmond	Ferdinand	15	55	50	41	74	13	81	100	89	94	91	97	
470	469	90409	Hansen	Kayla	59	15	98	26	99	98	60	25	25	13	82	71	
471	470	57574	Norris	Dion	100	64	69	35	70	88	28	97	68	6	5	36	
472	471	70039	Randolph	Stacy	59	25	22	0	31	14	94	15	83	5	4	20	
473	472	7158	Jimenez	Graham	91	33	28	47	77	93	19	70	50	20	7	18	
474	473	92015	Sherman	Kennith	21	46	93	97	60	99	11	28	75	9	17	97	
475	474	31739	Foley	Isabel	1	70	35	54	11	76	4	7	82	32	78	12	
476	475	66927	Guerrero	Billie	43	26	69	96	6	18	84	79	56	55	84	61	
477	476	36158	Ramsey	Myrna	100	90	48	85	67	96	37	13	49	94	87	28	
478	477	17831	Wyatt	Mickey	17	2	35	65	59	51	34	50	78	87	80	86	
479	478	66550	Gordon	Eula	70	54	1	37	40	67	60	46	85	57	54	37	
480	479	55564	Shaw	Earl	15	78	22	1	55	90	70	99	47	63	28	91	
481	480	34588	Faulkner	Gabriela	18	34	69	36	63	76	4	78	16	36	70	11	
482	481	98375	Wolf	Tod	76	66	22	74	94	26	100	87	29	54	90	21	
483	482	38851	Horton	Hai	55	34	23	28	9	86	71	22	73	85	79	21	
484	483	22475	Brooks	Cyrus	59	22	50	23	54	29	73	85	45	57	59	57	
485	484	53345	James	Dillon	52	5	92	30	0	95	43	93	29	2	25	89	

Sorting Data

Clicking sort gets you this

The screenshot displays the Microsoft Excel interface with the 'Data' tab selected. A data table is visible, and the 'Sort' dialog box is open, showing the 'Sort by' dropdown set to 'Cell Values' and the 'Order' dropdown set to 'A to Z'. The dialog box also includes options for 'Add Level', 'Delete Level', 'Copy Level', and 'Options...'. The 'My data has headers' checkbox is checked.

1	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q
	Number	Student ID	Last Name	First Name	Quiz 1	Quiz 2	Quiz 3	Quiz 4	Quiz 5	Exam 1	Exam 2	Exam 3	Homework 1	Homework 2	Homework 3	Homework 4	Homework 5
466	465	54980	Contreras	Malinda	66	5	60	8	5	25	77	72	99	62	26	76	
467	466	70180	Carlson	Columbus	33	40	25	54	40	23	63	66	66	74	45	97	
468	467	9224	Coleman	Cedric	21									45	83	59	
469	468	51552	Richmond	Ferdinand	15									94	91	97	
470	469	90409	Hansen	Kayla	59									13	82	71	
471	470	57574	Norris	Dion	100									6	5	36	
472	471	70039	Randolph	Stacy	59									5	4	20	
473	472	7158	Jimenez	Graham	91									20	7	18	
474	473	92015	Sherman	Kennith	21									9	17	97	
475	474	31739	Foley	Isabel	1									32	78	12	
476	475	66927	Guerrero	Billie	43									55	84	61	
477	476	36158	Ramsey	Myrna	100									94	87	28	
478	477	17831	Wyatt	Mickey	17									87	80	86	
479	478	66550	Gordon	Eula	70									57	54	37	
480	479	55564	Shaw	Earl	15	78	22	1	55	90	70	99	47	63	28	91	
481	480	34588	Faulkner	Gabriela	18	34	69	36	63	76	4	78	16	36	70	11	
482	481	98375	Wolf	Tod	76	66	22	74	94	26	100	87	29	54	90	21	
483	482	38851	Horton	Hai	55	34	23	28	9	86	71	22	73	85	79	21	
484	483	22475	Brooks	Cyrus	59	22	50	23	54	29	73	85	45	57	59	57	
485	484	53345	James	Dillon	52	5	92	30	0	95	43	93	29	2	25	89	

Sorting Data

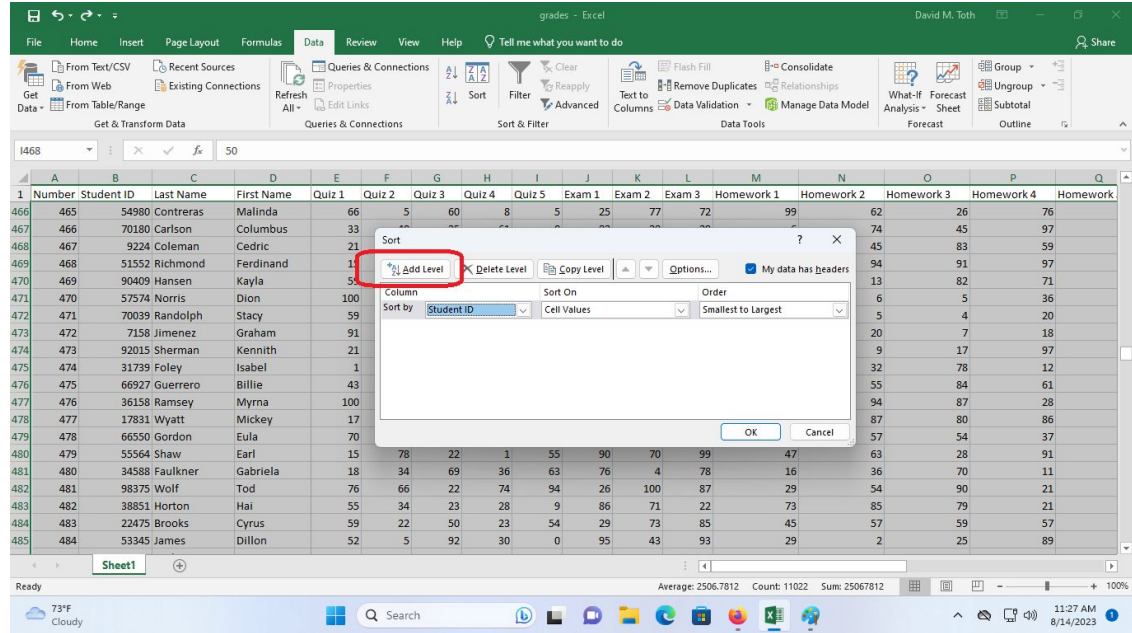
Select the column to sort by

The screenshot shows a Microsoft Excel spreadsheet titled "grades - Excel" with a data table containing student information and scores. The "Data" tab is active, and the "Sort" button in the "Data Tools" group is highlighted. The "Sort" dialog box is open, showing the "Sort by" dropdown menu with "Student ID" selected. The "Sort On" dropdown is set to "All Values" and the "Order" is "A to Z".

Number	Student ID	Last Name	First Name	Quiz 1	Quiz 2	Quiz 3	Quiz 4	Quiz 5	Exam 1	Exam 2	Exam 3	Homework 1	Homework 2	Homework 3	Homework 4	Homework 5
465	54980	Contreras	Malinda	66	5	60	8	5	25	77	72	99	62	26	76	
466	70180	Carlson	Columbus	33										45	97	
467	9224	Coleman	Cedric	21										83	59	
468	51552	Richmond	Ferdinand	15										91	97	
469	90409	Hansen	Kayla	59										82	71	
470	57574	Norris	Dion	100										5	36	
471	70039	Randolph	Stacy	59										4	20	
472	7158	Jimenez	Graham	91										7	18	
473	92015	Sherman	Kenneth	21										17	97	
474	31739	Foley	Isabel	1										78	12	
475	66927	Guerrero	Billie	43										84	61	
476	36158	Ramsey	Myrna	100										87	28	
477	17831	Wyatt	Mickey	17										80	86	
478	66550	Gordon	Eula	70										54	37	
479	55564	Shaw	Earl	15	78	22	1	55	90	70	99	47	63	28	91	
480	34588	Faulkner	Gabriela	18	34	69	36	63	76	4	78	16	36	70	11	
481	98375	Wolf	Tod	76	66	22	74	94	26	100	87	29	54	90	21	
482	38851	Horton	Hai	55	34	23	28	9	86	71	22	73	85	79	21	
483	22475	Brooks	Cyrus	59	22	50	23	54	29	73	85	45	57	59	57	
484	53345	James	Dillon	52	5	92	30	0	95	43	93	29	2	25	89	

Sorting Data

Add another
column to sort



The screenshot displays the Microsoft Excel interface with a data table. The table has columns for student information (Number, Student ID, Last Name, First Name) and quiz scores (Quiz 1 through Quiz 5, Exam 1 through Exam 3, Homework 1 through Homework 4). A 'Sort' dialog box is open, showing the 'Add Level' button highlighted with a red rectangle. The dialog box also shows the 'Sort by' dropdown set to 'Student ID' and the 'Sort On' dropdown set to 'Cell Values'. The 'Order' dropdown is set to 'Smallest to Largest'. The 'My data has headers' checkbox is checked.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q
1	Number	Student ID	Last Name	First Name	Quiz 1	Quiz 2	Quiz 3	Quiz 4	Quiz 5	Exam 1	Exam 2	Exam 3	Homework 1	Homework 2	Homework 3	Homework 4	Homework
466	465	54980	Contreras	Malinda	66	5	60	8	5	25	77	72	99	62	26	76	
467	466	70180	Carlson	Columbus	33									74	45	97	
468	467	9224	Coleman	Cedric	21									45	83	59	
469	468	51552	Richmond	Ferdinand	1									94	91	97	
470	469	90409	Hansen	Kayla	5									13	82	71	
471	470	57574	Norris	Dion	100									6	5	36	
472	471	70039	Randolph	Stacy	59									5	4	20	
473	472	7158	Jimenez	Graham	91									20	7	18	
474	473	92015	Sherman	Kennith	21									9	17	97	
475	474	31739	Foley	Isabel	1									32	78	12	
476	475	66927	Guerrero	Billie	43									55	84	61	
477	476	36158	Ramsey	Myrna	100									94	87	28	
478	477	17831	Wyatt	Mickey	17									87	80	86	
479	478	66550	Gordon	Eula	70									57	54	37	
480	479	55564	Shaw	Earl	15	78	22	1	55	90	70	99	47	63	28	91	
481	480	34588	Faulkner	Gabriela	18	34	69	36	63	76	4	78	16	36	70	11	
482	481	98375	Wolf	Tod	76	66	22	74	94	26	100	87	29	54	90	21	
483	482	38851	Horton	Hai	55	34	23	28	9	86	71	22	73	85	79	21	
484	483	22475	Brooks	Cyrus	59	22	50	23	54	29	73	85	45	57	59	57	
485	484	53345	James	Dillon	52	5	92	30	0	95	43	93	29	2	25	89	

Sorting Data

The screenshot displays the Microsoft Excel interface with a spreadsheet titled "grades - Excel". The "Data" tab is active, showing various data management tools. A "Sort" dialog box is open, indicating a sort operation is being configured. The spreadsheet data is as follows:

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q
1	Number	Student ID	Last Name	First Name	Quiz 1	Quiz 2	Quiz 3	Quiz 4	Quiz 5	Exam 1	Exam 2	Exam 3	Homework 1	Homework 2	Homework 3	Homework 4	Homework 5
466	465	54980	Contreras	Malinda	66	5	60	8	5	25	77	72	99	62	26	76	
467	466	70180	Carlson	Columbus	33	10	25	74	28	88	88	80	74	45	97		
468	467	9224	Coleman	Cedric	21								45	83	59		
469	468	51552	Richmond	Ferdinand	15								94	91	97		
470	469	90409	Hansen	Kayla	59								13	82	71		
471	470	57574	Norris	Dion	100								6	5	36		
472	471	70039	Randolph	Stacy	59								5	4	20		
473	472	7158	Jimenez	Graham	91								20	7	18		
474	473	92015	Sherman	Kennith	21								32	17	97		
475	474	31739	Foley	Isabel	1								32	78	12		
476	475	66927	Guerrero	Billie	43								55	84	61		
477	476	36158	Ramsey	Myrna	100								94	87	28		
478	477	17831	Wyatt	Mickey	17								87	80	86		
479	478	66550	Gordon	Eula	70								57	54	37		
480	479	55564	Shaw	Earl	15	78	22	1	55	90	70	99	47	63	28	91	
481	480	34588	Faulkner	Gabriela	18	34	69	36	63	76	4	78	16	36	70	11	
482	481	98375	Wolf	Tod	76	66	22	74	94	26	100	87	29	54	90	21	
483	482	38851	Horton	Hai	55	34	23	28	9	86	71	22	73	85	79	21	
484	483	22475	Brooks	Cyrus	59	22	50	23	54	29	73	85	45	57	59	57	
485	484	53345	James	Dillon	52	5	92	30	0	95	43	93	29	2	25	89	

The "Sort" dialog box is configured with the following settings:

- Column: Student ID
- Sort by: Student ID
- Sort On: Cell Values
- Order: Smallest to Largest
- Then by: Last Name
- Sort On: Cell Values
- Order: A to Z
- My data has headers: ☒

The status bar at the bottom shows "Ready", "Average: 2506.7812", "Count: 11022", "Sum: 25067812", and the system clock "11:27 AM 8/14/2023".

Making Charts

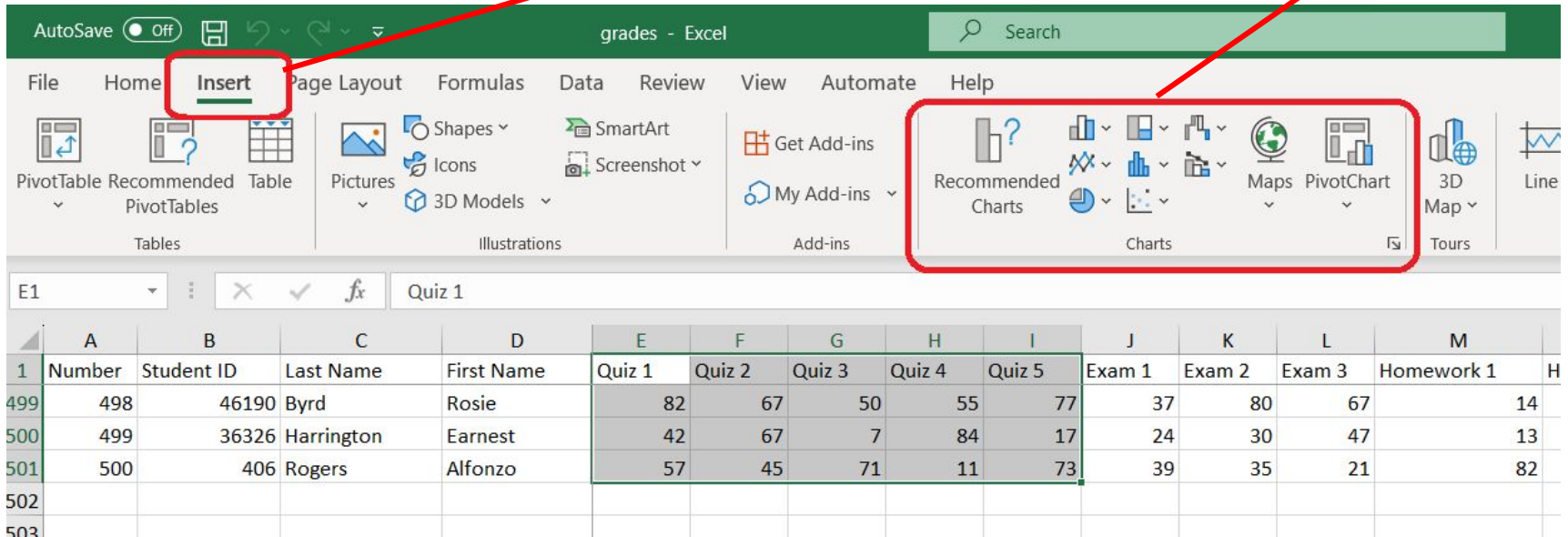
DSC 280 Data Visualization will talk about what charts to use & why.

We're going to focus on

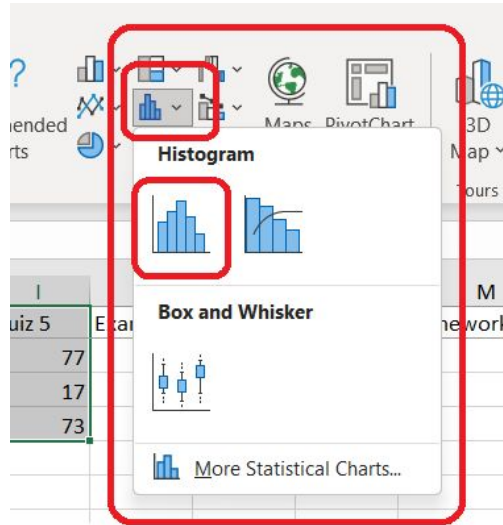
- Histograms: show frequency of something
- Bar graphs: allow comparison of values across subgroups of data
- Line graphs: illustrate trends over time
- Scatter plots: can show relationship between 2 variables
- Pie charts: shows values as percentage of whole - **only use this when somebody demands it**

Making Charts

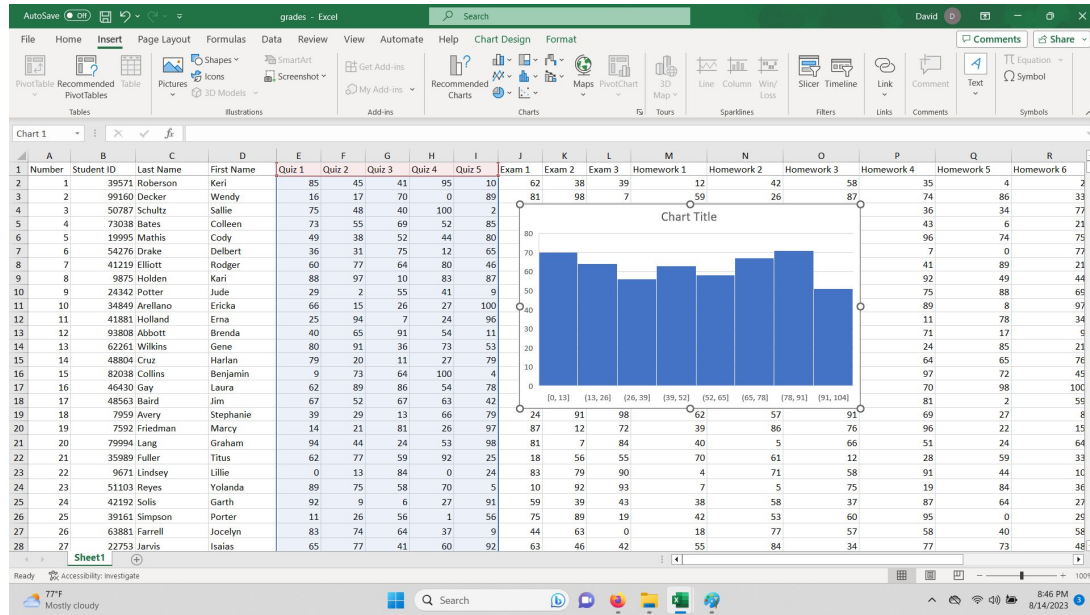
Select your data, go to the Insert tab, and choose the appropriate graph type



Histogram



Histogram

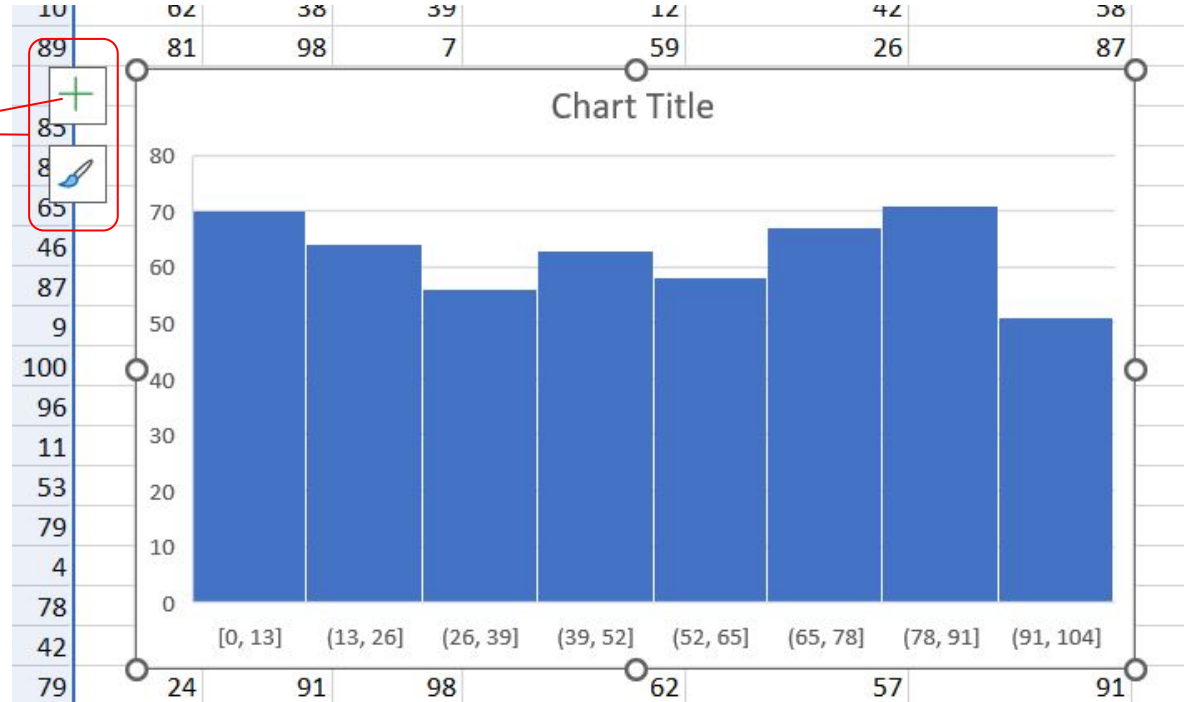


Histogram

Double click the histogram
to bring up these

Use + to change

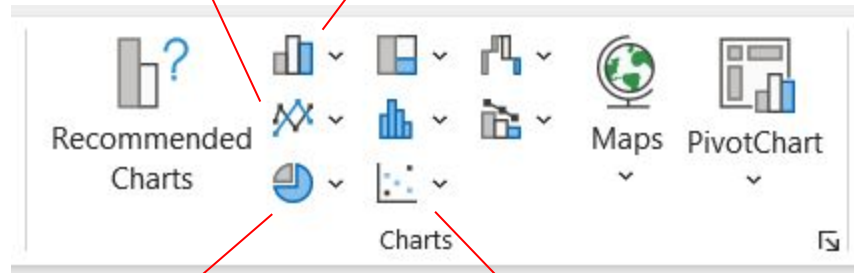
- Titles
- Axes
- Data Labels
- Gridlines
- Legend



Other Graphs

Line graph

Bar graph



Pie chart

Scatter plot

Lab Time!

Download fastfood.xlsx from Moodle

Lab handout:

https://docs.google.com/document/d/1SI2gPy8Jq_hpiJ2B4eSz87lvk99dUMNN71s_xr5aDXU/edit?usp=sharing

Excel Day 2

Day 2 Topics

Renaming Cells

COUNT

Importing & Exporting Data

Cleaning Data

Converting Data

Remove duplicate rows

UNIQUE

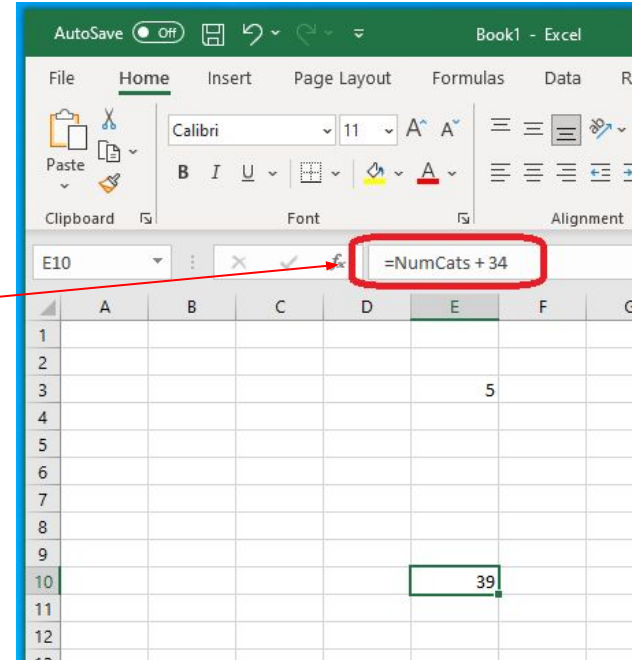
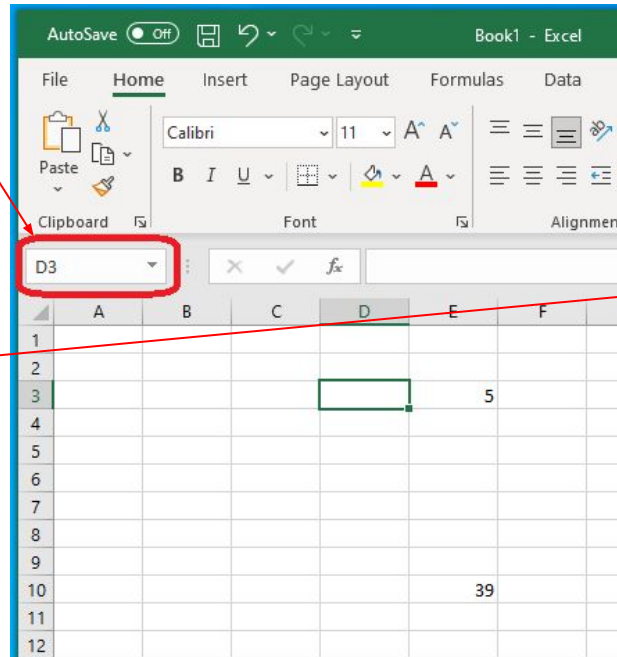
VLOOKUP

Useful built-in functions

Renaming Cells

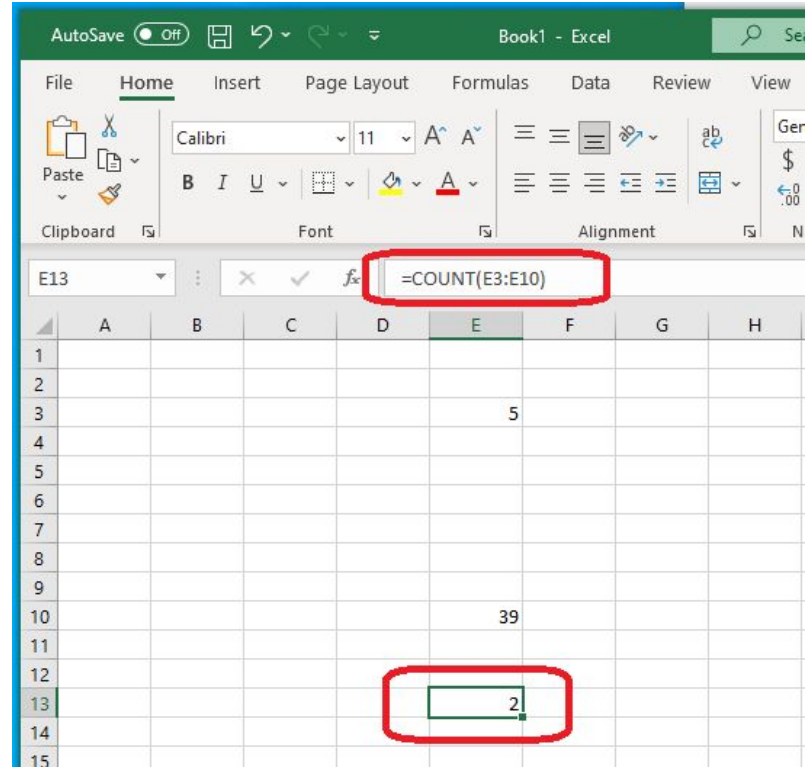
Click dropdown,
type in desired
name, hit Enter key

Can now use name
in formula



COUNT

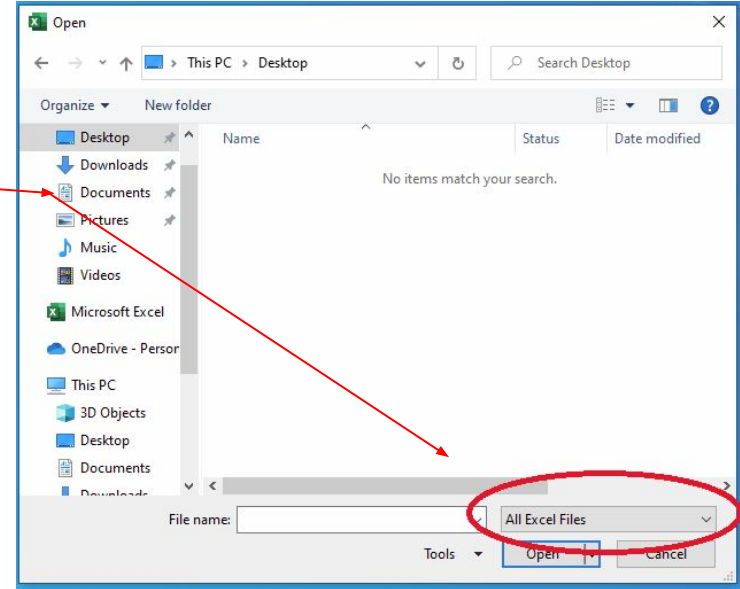
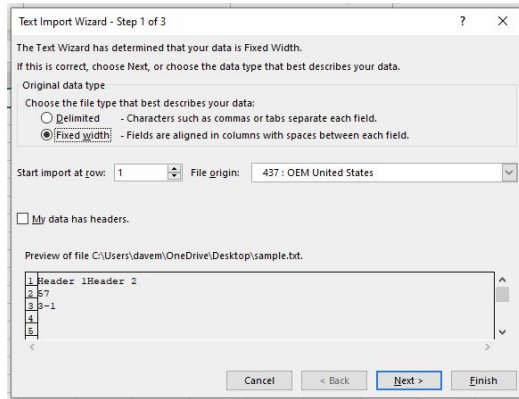
Count how many cells in a range have entries in them



Importing Data From Text Files

Opening them: File->Open->

Change drop-down to All Files,
navigate to file and left click and Open



Importing Data From Text Files

Text Import Wizard - Step 1 of 3

The Text Wizard has determined that your data is Fixed Width.
If this is correct, choose Next, or choose the data type that best describes your data.

Original data type

Choose the file type that best describes your data:

☐ Delimited - Characters such as commas or tabs separate each field.

☒ Fixed width - Fields are aligned in columns with spaces between each field.

Start import at row: 1 File origin: 437 : OEM United States

☐ My data has headers.

Preview of file C:\Users\davem\OneDrive\Desktop\sample.txt.

1	Header 1Header 2
2	57
3	3-1
4	
5	

Cancel < Back Next > Finish

1 - Select Delimited vs Fixed width.
We'll choose Delimited for this example.

2 - Click Next.

Importing Data From Text Files

Text Import Wizard - Step 2 of 3

This screen lets you set the delimiters your data contains. You can see how your text is affected in the preview below.

Delimiters

☒ Tab
☐ Semicolon
☐ Comma
☐ Space
☐ Other:

☐ Treat consecutive delimiters as one

Text qualifier: ▼

Data preview

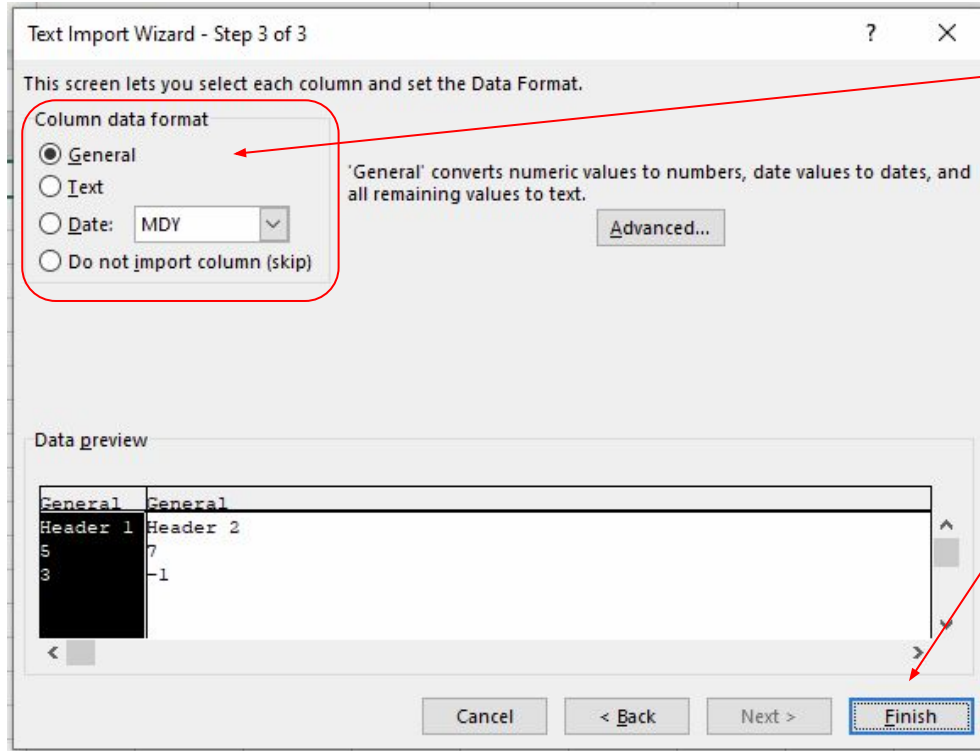
Header 1	Header 2
5	7
3	-1

Cancel < Back **Next >** Finish

1 - Choose the delimiter(s).
We'll choose Tab for this example.

2 - Click Next.

Importing Data From Text Files



1 - Choose the data format.
We'll choose General for this example.

2 - Click Finish.

3 - Voila!

	A	B	C	D
1	Header 1	Header 2		
2	5	7		
3	3	-1		
4				

Importing Data From CSV Files

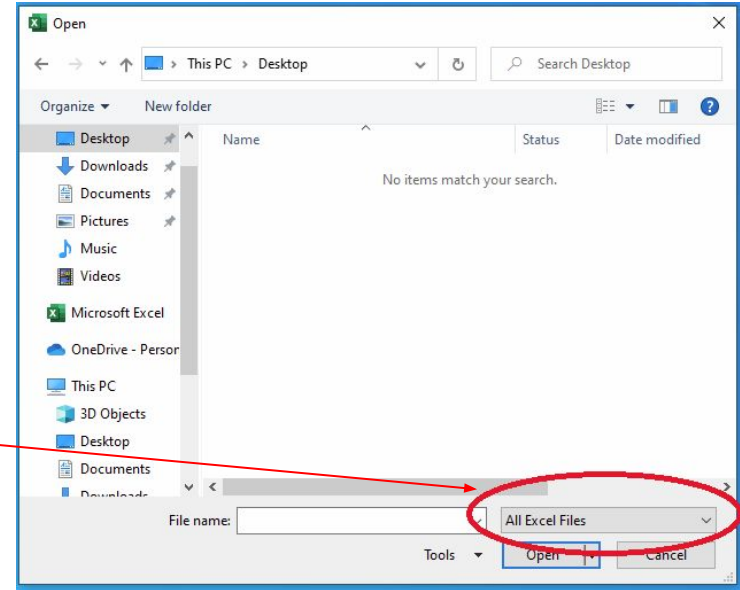
What are CSV files?

Beware CSV files w/ commas in addresses, etc!

Opening them: File->Open->

Change drop-down to All Files,
navigate to file and left click
and Open

Can also just double-click to open



Summing Data (Review)

Left click the cell to hold the sum

Type =sum(*sum start cell:sum stop cell*) and hit the Enter key

OR

Type =sum(and then left click & hold and drag to select range and then hit Enter key)

Can also have comma separated list of cells to sum inside parens instead

Motivation: Open first_broken_numbers.xlsx

Do a sum on columns C and D.

WTF?

(see next slides, trying techniques!)

Weird Character Problems

ASCII - American Standard
Code for Information Interchange

ASCII control characters			ASCII printable characters			Extended ASCII characters										
00	NULL	(Null character)	32	space	64	@	96	`	128	Ç	160	á	192	Ł	224	Ó
01	SOH	(Start of Header)	33	!	65	A	97	a	129	ù	161	í	193	ł	225	ô
02	STX	(Start of Text)	34	"	66	B	98	b	130	é	162	ó	194	Ł	226	õ
03	ETX	(End of Text)	35	#	67	C	99	c	131	â	163	ú	195	ł	227	ö
04	EOT	(End of Trans.)	36	\$	68	D	100	d	132	ä	164	ñ	196	—	228	ø
05	ENQ	(Enquiry)	37	%	69	E	101	e	133	å	165	Ň	197	†	229	ő
06	ACK	(Acknowledgement)	38	&	70	F	102	f	134	â	166	ª	198	‡	230	û
07	BEL	(Bell)	39	'	71	G	103	g	135	ç	167	º	199	Ä	231	ü
08	BS	(Backspace)	40	(	72	H	104	h	136	ê	168	¿	200	ℒ	232	þ
09	HT	(Horizontal Tab)	41	)	73	I	105	i	137	ë	169	®	201	ℓ	233	Û
10	LF	(Line feed)	42	*	74	J	106	j	138	è	170	¬	202	ℓ	234	Ü
11	VT	(Vertical Tab)	43	+	75	K	107	k	139	ï	171	½	203	ℓ	235	Ù
12	FF	(Form feed)	44	,	76	L	108	l	140	î	172	¼	204	ℓ	236	ý
13	CR	(Carriage return)	45	-	77	M	109	m	141	í	173	⅓	205	ℓ	237	Ÿ
14	SO	(Shift Out)	46	.	78	N	110	n	142	Ä	174	«	206	ℓ	238	ˆ
15	SI	(Shift In)	47	/	79	O	111	o	143	Å	175	»	207	ℓ	239	˜
16	DLE	(Data link escape)	48	0	80	P	112	p	144	É	176	⌘	208	ð	240	≡
17	DC1	(Device control 1)	49	1	81	Q	113	q	145	æ	177	⌘	209	Ð	241	±
18	DC2	(Device control 2)	50	2	82	R	114	r	146	Æ	178	⌘	210	É	242	≡
19	DC3	(Device control 3)	51	3	83	S	115	s	147	ó	179	⌘	211	Ê	243	¼
20	DC4	(Device control 4)	52	4	84	T	116	t	148	ô	180	⌘	212	Ë	244	½
21	NAK	(Negative acknowl.)	53	5	85	U	117	u	149	õ	181	⌘	213	Ì	245	¾
22	SYN	(Synchronous idle)	54	6	86	V	118	v	150	ù	182	⌘	214	Í	246	+
23	ETB	(End of trans. block)	55	7	87	W	119	w	151	ú	183	⌘	215	Î	247	°
24	CAN	(Cancel)	56	8	88	X	120	x	152	ÿ	184	©	216	Ï	248	ˆ
25	EM	(End of medium)	57	9	89	Y	121	y	153	Ö	185	⌘	217	⌘	249	˜
26	SUB	(Substitute)	58	:	90	Z	122	z	154	Ü	186	⌘	218	⌘	250	·
27	ESC	(Escape)	59	;	91	[	123	{	155	ø	187	⌘	219	⌘	251	ˆ
28	FS	(File separator)	60	<	92	\	124		156	£	188	⌘	220	⌘	252	˜
29	GS	(Group separator)	61	=	93	]	125	}	157	Ø	189	¢	221	⌘	253	ˆ
30	RS	(Record separator)	62	>	94	^	126	~	158	×	190	¥	222	⌘	254	˜
31	US	(Unit separator)	63	?	95	_			159	f	191	γ	223	⌘	255	nbsp
127	DEL	(Delete)														

<https://theasciicode.com.ar/>

Trimming Off Unwanted Characters

Why? Won't behave right (SUM doesn't work, etc.)

TRIM deletes space character (ASCII 32) - not other whitespace

=TRIM(some_cell)

CLEAN deletes first 32 non-printing characters in ASCII (0-31)

=CLEAN(some_cell)

Can combine

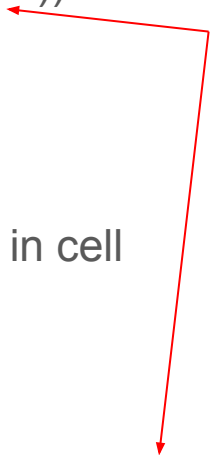
=TRIM(CLEAN(some_cell))

Trimming Off Unwanted Characters

What about other characters like 160 ()?

=TRIM(SUBSTITUTE(some_cell, CHAR(160), " "))

Don't copy-paste from the slide or the quotes may cause an issue!



How do I know what the offending character is?

=CODE(some_cell) gets code for 1st character in cell

Find all characters in cell B3

=IFERROR(TEXTJOIN(",", TRUE, CODE(MID(B3, SEQUENCE(LEN(B3)), 1))), "EMPTY")

Trimming Off Unwanted Characters

Trimmed data winds up in new cell(s)/column

I want the data in its original location!

Copy the new cells

Paste special (values) over old cells

Right click and select



Converting Data

What if your sum is wrong? (show example)

You might have strings, not numbers

Excel gives a visual indication of this!



How do we get numbers?

Put a 1 in a cell, copy it, select the cells to fix, do Paste-special Multiply (only desktop version)

OR

=VALUE(some_cell)

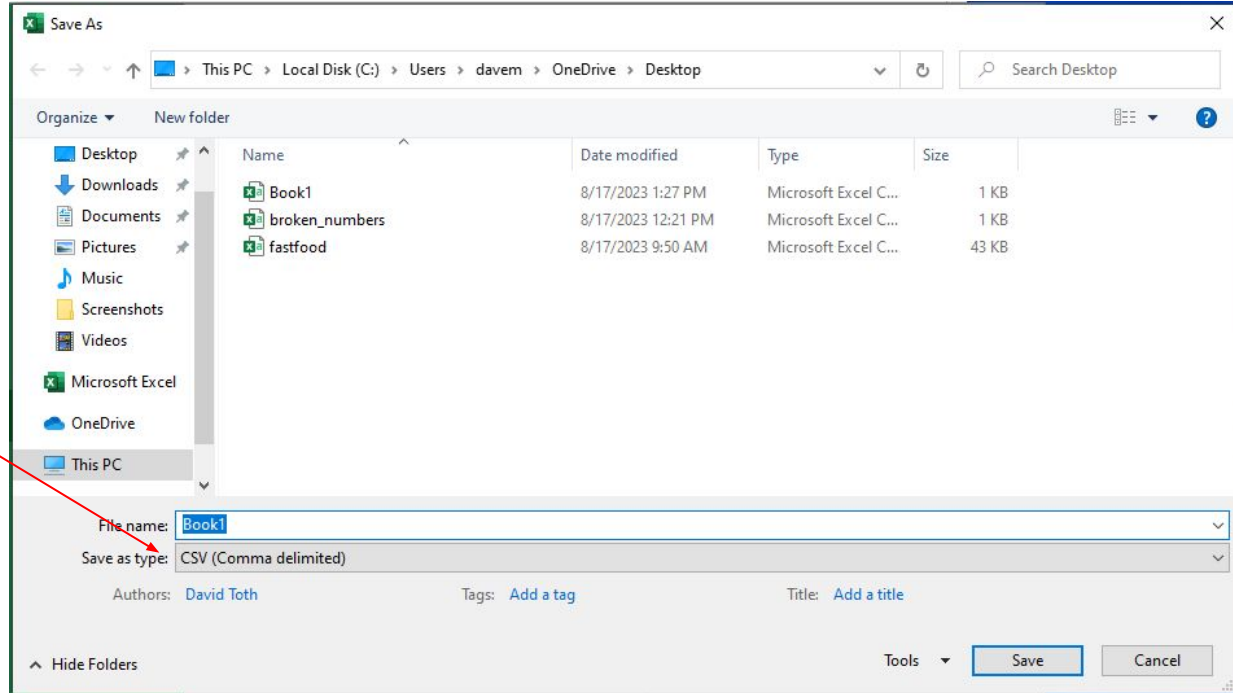
But you may have to get rid of the offending character(s) with SUBSTITUTE, TRIM, or CLEAN before taking the VALUE of it.



Exporting Data

File->Save As

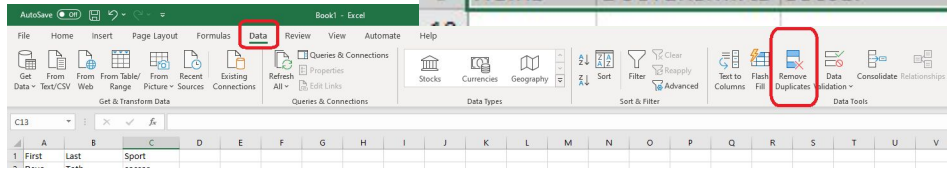
Select the type
from the
dropdown



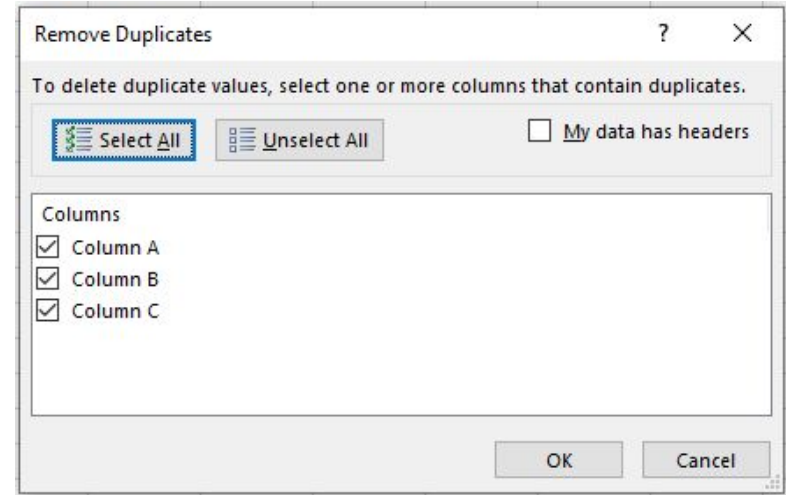
Removing Duplicate Rows

Select your data
From Data tab,
click Remove
Duplicates

	A	B	C
1	First	Last	Sport
2	Dave	Toth	soccer
3	ImAn	Example	tennis
4	IRun	Fast	track
5	Ima	Doofus	soccer
6	Heinz	Doofensmirtz	soccer
7	IRun	Fast	cross country
8	IRun	Fast	track
9	Heinz	Doofensmirtz	soccer



Select the columns and click OK - note that
the columns selected has an impact!!!



UNIQUE

Gets you the unique values...

Vanilla form - there are optional arguments (later)

[illegible]

VLOOKUP

Vertical lookup - helps you look for a specified value by searching for it vertically across the sheet

The image shows an Excel spreadsheet with a VLOOKUP formula in cell H9. The formula is `=VLOOKUP(H9,A6:F15,2,FALSE)`. A red box highlights the formula bar. A red arrow points from the formula bar to cell H9, which contains the result 'HP'. Another red arrow points from cell H9 to cell G9, which contains the lookup value '3'. A green arrow points from the number '2' in the formula to column B, which is labeled 'Result Column'. A red arrow points from the number '1' in the formula to column A, which is labeled 'Lookup column'. The data table is as follows:

ID	Company	Product	TypeName	Ram	Price_euros
1	Apple	MacBook Pro	Ultrabook	8	1339.69
2	Lenovo	IdeaPad 320-15ISK	Notebook	4	369
3	HP	ProBook 440	Notebook	8	860
4	Microsoft	Surface Laptop	Ultrabook	4	1089
5	Dell	Inspiron 3567	Notebook	4	498.9
6	Acer	Aspire 3	Notebook	4	400
7	Apple	Macbook Air	Ultrabook	8	1158.7
8	Asus	ZenBook	Ultrabook	16	1495
9	Acer	Swift 3	Ultrabook	8	770
10	HP	Pavilion 15	Ultrabook	8	699

Below the data table, there is a small table with two columns: 'Company ID' and 'Company'. The first row contains the values '3' and 'HP' respectively. Red arrows point from the '3' in the 'Company ID' column to the '3' in the formula, and from the 'HP' in the 'Company' column to the 'HP' in cell H9.

<https://www.simplilearn.com/tutorials/excel-tutorial/vlookup-in-excel>

VLOOKUP Syntax

```
=VLOOKUP(lookup_value, table_array, col_index_number,[range_lookup])
```

Where,

- **lookup_value:** This specifies the value that you want to look up in our data.
- **table_array:** This is the location where the values are present in excel.
- **col_index_number:** This specifies the column number from where we need to return the value.
- **range_lookup:** This has two options; if the value is set to FALSE, that means we are looking for an exact match. If the value is TRUE, then we are looking for an approximate match.

Useful Built-In Functions

- The following functions are particularly helpful for use in manipulating data:

Function Table	
Function	Description
= Average(A1:A10)	Calculates the average of a set of data
= Count(A1:A10)	Counts the number of cells in a range that contain numbers
= CountA(A1:A10)	Counts the number of cells in a range that are not empty
= Countif(A1:A10,">10")	Counts the number of cells that meet a certain condition
= Large(A1:A10,n)	Returns the nth largest value in a data set
= Max(A1:A10)	Finds the maximum value in a set of data
= Median(A1:A10)	Finds the median value in a set of data
= Min(A1:A10)	Finds the minimum value in a set of data
= Mround(A1,5)	Returns a number rounded to the desired multiple
= Product(A1:A10)	Multiplies a range of data
= Rand()	Returns a random number between 0 and 1
= Randbetween(bottom,top)	Returns a random number between the numbers specified
= Round(A1,2)	Rounds a selected cell to a specified number of digits
= Small(A1:A10,n)	Returns the nth smallest value in a data set
= Sum(A1:A10)	Sums a range of data
= Sumproduct(A1:A10,B1:B10)	Sums the Products of pairs of data
= Sumif(A1:A10,">10")	Sums a set of data that meet selected criteria

Lab Time!

Lab activity is linked from Moodle.

Excel Day 3

Day 3 Topics

Filtering

Grouping

Transposing Data

Date Functions

String Manipulation (LEFT, RIGHT, CONCATENATE)

Splitting columns

Filtering

Allows you to temporarily hide data to focus on the portion you want to

How to Filter

Select a cell in the data

Click the Filter button

You'll get column
header arrows

Click an arrow to
get...

The screenshot shows the Microsoft Excel interface with the 'Data' tab selected in the ribbon. The 'Filter' button, represented by a funnel icon, is highlighted in the 'Sort & Filter' group. A red arrow points from the 'Filter' button to the 'calories' column header in the data table below.

	A	B	C	D	E	F	G	H	I	J	K	L	M
	restaurant	item	calories	cal_fat	total_fat	sat_fat	trans_fat	cholesterol	sodium	total_carb	fiber	sugar	protein
2	Sonic	5 Piece Crispy Chicken Tender Dinne	470	220	24	4.5	0	0	1340	26	0	0	
3	Taco Bell	Cheesy Bean and Rice Burrito	420	160	17	3.5	0	0	930	55	6	4	
4	Taco Bell	Fresco Bean Burrito	350	80	9	3	0	0	950	57	9	3	
5	Sonic	3 Piece Crispy Chicken Tender Dinne	280	130	14	2.5	0	0	800	16	0	0	
6	Subway	Footlong Veggie Delite	460	40	6	2	0	0	620	88	10	12	
7	Taco Bell	Triple Layer Nachos	320	140	15	1.5	0	0	600	41	6	2	
8	Domino's	Branded Meatballs	250	20	2	1	0	0	500	25	2	1	

How to Filter

Clicking column header arrow gets

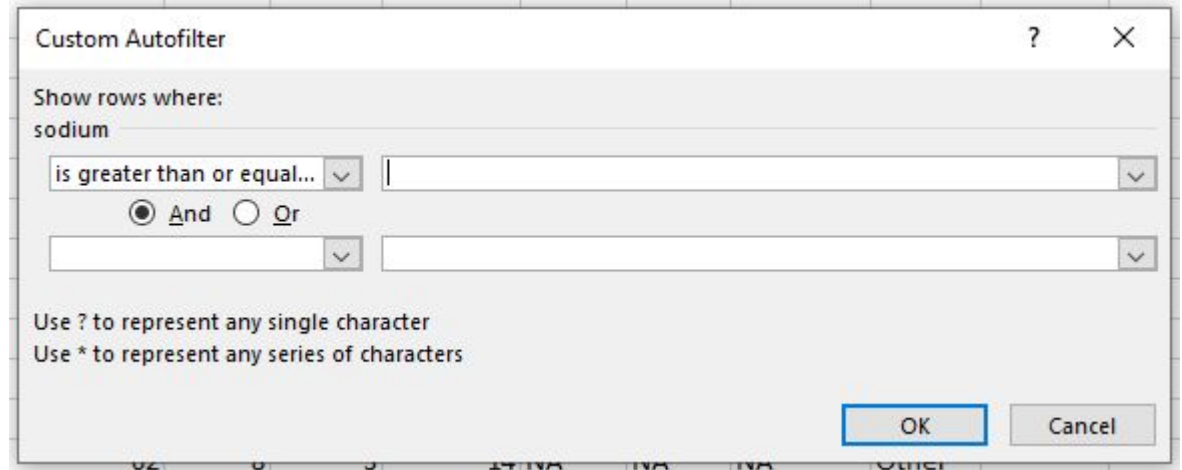
Pick your option

The screenshot shows a spreadsheet with columns F through L. The 'sodium' column (I) has a dropdown arrow. The menu is open, showing options like 'Sort Smallest to Largest', 'Sort Largest to Smallest', 'Sort by Color', 'Sheet View', 'Clear Filter From "sodium"', 'Filter by Color', and 'Number Filters'. The 'Number Filters' section is expanded, showing a search bar and a list of values with checkboxes. The 'Greater Than Or Equal To...' option is highlighted with a red box.

	F	G	H	I	J	K	L	pro
4	sat_fat	trans_fat	cholesterol	sodium	total_carb	fiber	sugar	
7					26	0	0	
9					55	6	4	
4					57	9	3	
6					16	0	0	
5					88	10	12	
9					41	6	2	
3					36	2	1	
7					44	5	6	
0								
2								
1								
1								
6								
5								
6								
4								
4								
4								
0								
2								
3								
1	3	0	10	1310				
0	2.5	0	10	440				
7	1	0	10	800				
2	0.5	0	10	450	20	2	5	

How to Filter

Fill in and click OK



The screenshot shows a 'Custom Autofilter' dialog box. At the top, it says 'Show rows where:' followed by the text 'sodium'. Below this, there are two rows of filter criteria. The first row has a dropdown menu set to 'is greater than or equal...' and an empty text input field. The second row has two empty text input fields. Between these two rows, there are two radio buttons: 'And' (which is selected) and 'Or'. At the bottom of the dialog, there are two lines of instructions: 'Use ? to represent any single character' and 'Use * to represent any series of characters'. In the bottom right corner, there are two buttons: 'OK' and 'Cancel'. The 'OK' button is highlighted with a blue border.

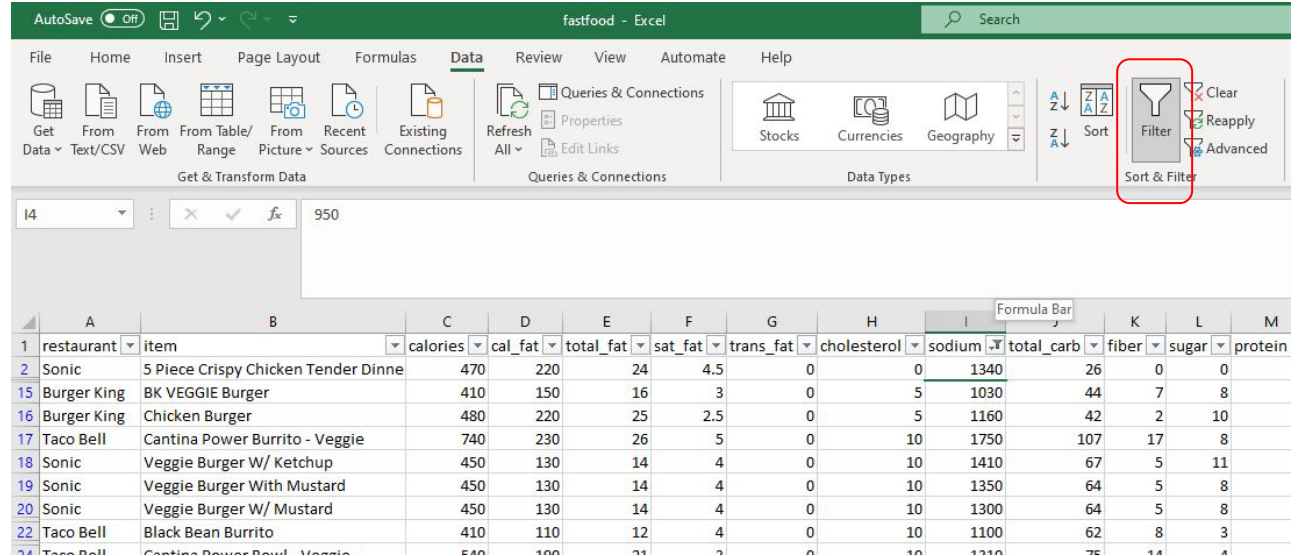
See the data now shown is only that matching your criteria

Clearing the Filter

Done with the filter?

Want to see all the data again?

Click the Filter button again to turn the filter off and see all the data.



The screenshot shows the Microsoft Excel interface with the 'Data' tab selected on the ribbon. The 'Filter' button, represented by a funnel icon, is highlighted with a red rectangle. To the right of the 'Filter' button are buttons for 'Clear', 'Reapply', and 'Advanced'. Below the ribbon, the spreadsheet is visible, showing a table of restaurant items. The table has columns for 'restaurant', 'item', 'calories', 'cal_fat', 'total_fat', 'sat_fat', 'trans_fat', 'cholesterol', 'sodium', 'total_carb', 'fiber', 'sugar', and 'protein'. The data is filtered to show only items with 'total_fat' greater than 20.

	A	B	C	D	E	F	G	H	I	J	K	L	M
	restaurant	item	calories	cal_fat	total_fat	sat_fat	trans_fat	cholesterol	sodium	total_carb	fiber	sugar	protein
2	Sonic	5 Piece Crispy Chicken Tender Dinne	470	220	24	4.5	0	0	1340	26	0	0	
15	Burger King	BK VEGGIE Burger	410	150	16	3	0	5	1030	44	7	8	
16	Burger King	Chicken Burger	480	220	25	2.5	0	5	1160	42	2	10	
17	Taco Bell	Cantina Power Burrito - Veggie	740	230	26	5	0	10	1750	107	17	8	
18	Sonic	Veggie Burger W/ Ketchup	450	130	14	4	0	10	1410	67	5	11	
19	Sonic	Veggie Burger With Mustard	450	130	14	4	0	10	1350	64	5	8	
20	Sonic	Veggie Burger W/ Mustard	450	130	14	4	0	10	1300	64	5	8	
22	Taco Bell	Black Bean Burrito	410	110	12	4	0	10	1100	62	8	3	
24	Taco Bell	Cantina Power Bowl - Veggie	540	180	21	3	0	10	1210	75	14	4	

Can I Sort Filtered Data?

Yup. Combine all the fun stuff you learned with filtering for even more power!

Grouping

Why?

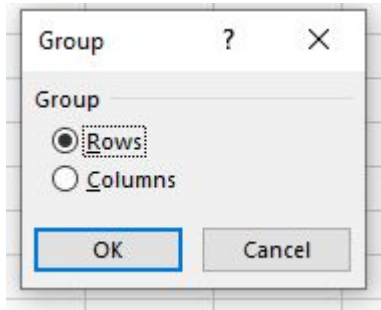
- Organize doc for efficiency
- Allow us to easily show/hide data
- Create summaries

Grouping Columns

Select columns to group (must be a continuous range)

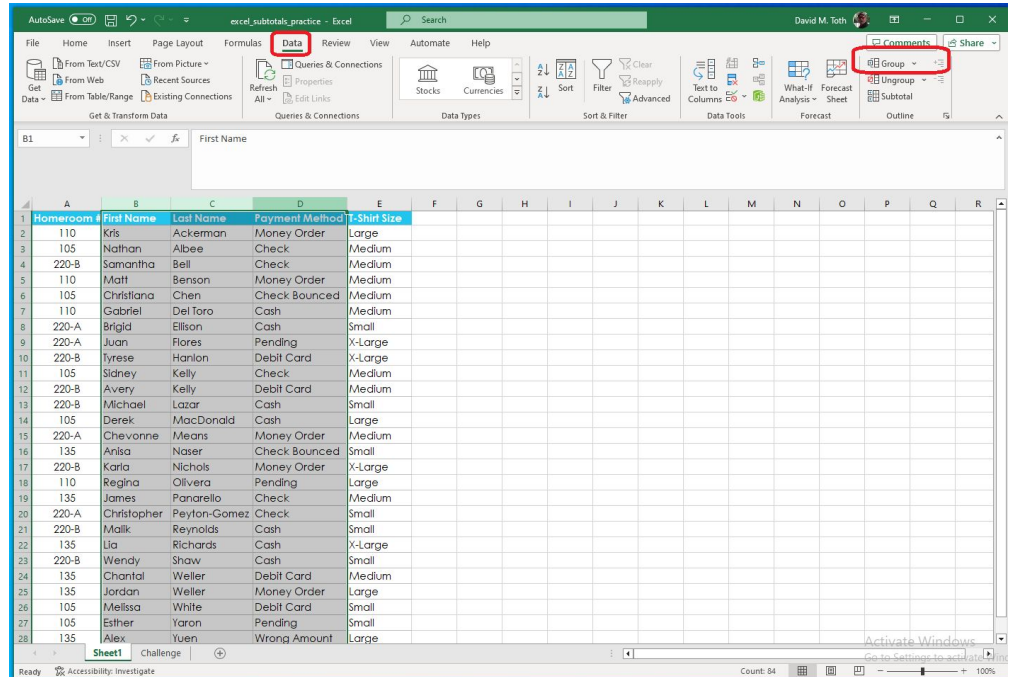
Click Group from the Data tab

Select rows or columns



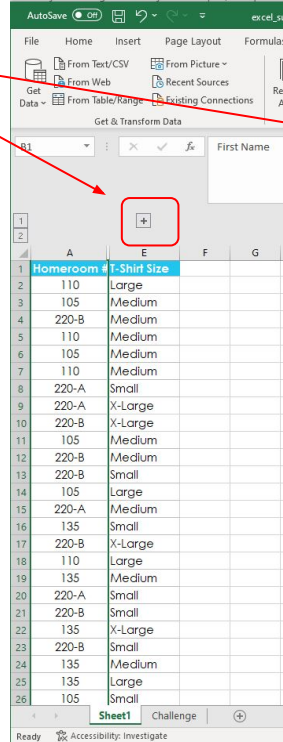
Data from

<https://edu.gcfglobal.org/en/excel/groups-and-subtotals/1/>



Grouping

Can hide/show with



excel_sub

File Home Insert Page Layout Formulas

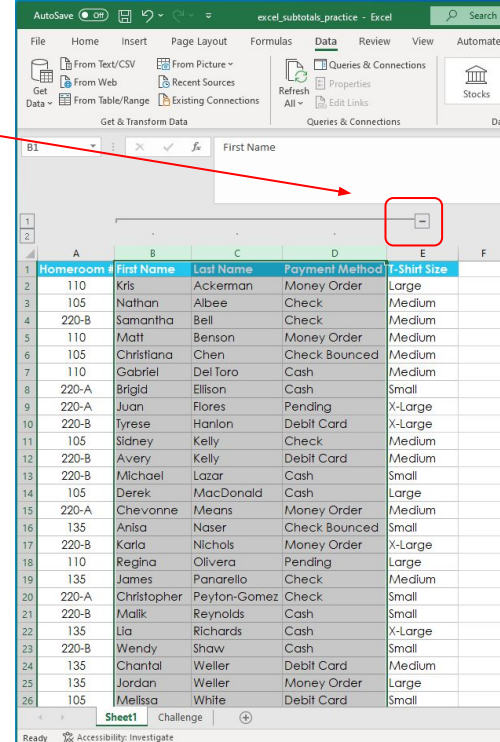
Get Data From Text/CSV From Picture From Web Recent Sources From Table/Range Existing Connections

Get & Transform Data

B1 First Name

	A	B	C	D	E	F	G
1	Homeroom #	T-Shirt Size					
2	110	Large					
3	105	Medium					
4	220-B	Medium					
5	110	Medium					
6	105	Medium					
7	110	Medium					
8	220-A	Small					
9	220-A	X-Large					
10	220-B	X-Large					
11	105	Medium					
12	220-B	Medium					
13	220-B	Small					
14	105	Large					
15	220-A	Medium					
16	135	Small					
17	220-B	X-Large					
18	110	Large					
19	135	Medium					
20	220-A	Small					
21	220-B	Small					
22	135	X-Large					
23	220-B	Small					
24	135	Medium					
25	135	Large					
26	105	Small					

Sheet1 Challenge



excel_subtotals_practice - Excel

File Home Insert Page Layout Formulas Data Review View Automate

Get Data From Text/CSV From Picture From Web Recent Sources From Table/Range Existing Connections

Get & Transform Data

B1 First Name

	A	B	C	D	E	F
1	Homeroom #	First Name	Last Name	Payment Method	T-Shirt Size	
2	110	Kris	Ackerman	Money Order	Large	
3	105	Nathan	Albee	Check	Medium	
4	220-B	Samantha	Bell	Check	Medium	
5	110	Matt	Benson	Money Order	Medium	
6	105	Christiana	Chen	Check Bounced	Medium	
7	110	Gabriel	Del Toro	Cash	Medium	
8	220-A	Brigid	Ellison	Cash	Small	
9	220-A	Juan	Flores	Pending	X-Large	
10	220-B	Tyrese	Hanlon	Debit Card	X-Large	
11	105	Sidney	Kelly	Check	Medium	
12	220-B	Avery	Kelly	Debit Card	Medium	
13	220-B	Michael	Lazar	Cash	Small	
14	105	Derek	MacDonald	Cash	Large	
15	220-A	Chevonne	Means	Money Order	Medium	
16	135	Anisa	Naser	Check Bounced	Small	
17	220-B	Karla	Nichols	Money Order	X-Large	
18	110	Regina	Olivera	Pending	Large	
19	135	James	Panarello	Check	Medium	
20	220-A	Christopher	Peyton-Gomez	Check	Small	
21	220-B	Malik	Reynolds	Cash	Small	
22	135	Lia	Richards	Cash	X-Large	
23	220-B	Wendy	Shaw	Cash	Small	
24	135	Chantal	Weller	Debit Card	Medium	
25	135	Jordan	Weller	Money Order	Large	
26	105	Melissa	White	Debit Card	Small	

Sheet1 Challenge

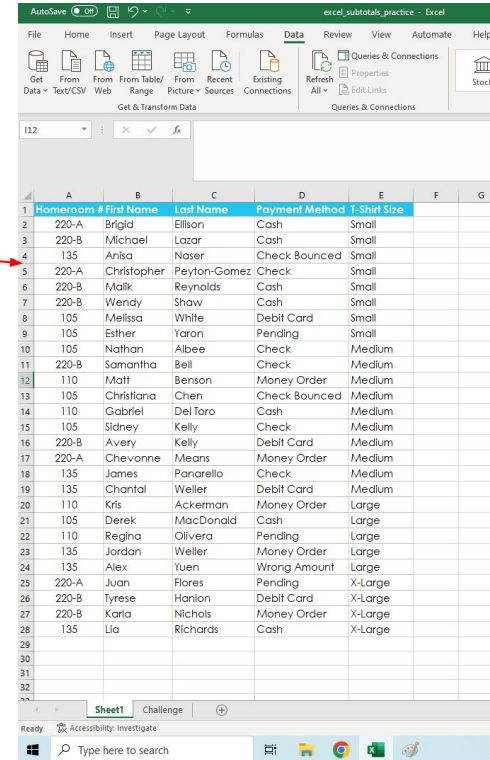
Grouping: Subtotals

Want a count, sum, avg of a set of things?

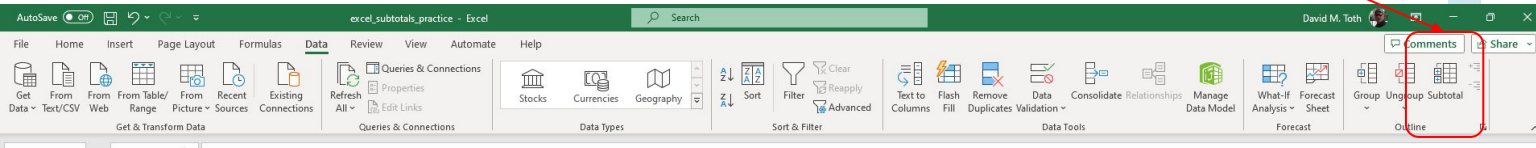
Sort by column you want the summary for

Click a cell in the data

Click Subtotal button

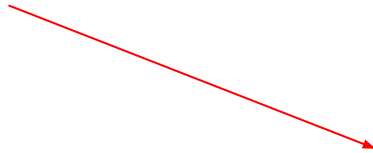


HomeRoom	#	First Name	Last Name	Payment Method	T-Shirt Size
220-A		Brigid	Ellison	Cash	Small
220-B		Michael	Lazar	Cash	Small
135		Anisa	Naser	Check Bounced	Small
220-A		Christopher	Peyton-Gomez	Check	Small
220-B		Malik	Reynolds	Cash	Small
105		Wendy	Shaw	Cash	Small
105		Melissa	White	Debit Card	Small
105		Esther	Yaron	Pending	Small
105		Nathan	Albee	Check	Medium
220-B		Samantha	Bell	Check	Medium
110		Matt	Benson	Money Order	Medium
105		Christiana	Chen	Check Bounced	Medium
110		Gabriel	Del Toro	Cash	Medium
105		Sidney	Kelly	Check	Medium
220-B		Avery	Kelly	Debit Card	Medium
220-A		Chevonne	Means	Money Order	Medium
135		James	Panarello	Check	Medium
135		Chantal	Weller	Debit Card	Medium
110		Kris	Ackerman	Money Order	Large
105		Derek	MacDonald	Cash	Large
110		Regina	Olivera	Pending	Large
135		Jordan	Weller	Money Order	Large
135		Alex	Yuen	Wrong Amount	Large
220-A		Juan	Flores	Pending	X-Large
220-B		Tyrese	Hanlon	Debit Card	X-Large
220-B		Karla	Nichols	Money Order	X-Large
135		Lia	Richards	Cash	X-Large



Grouping: Subtotals

Select what you need from the options



The 'Subtotal' dialog box is shown with the following settings:

- At each change in:** Homeroom #
- Use function:** Count
- Add subtotal to:**
 - ☐ Homeroom #
 - ☐ First Name
 - ☐ Last Name
 - ☐ Payment Method
 - ☒ T-Shirt Size
- ☒ Replace current subtotals
- ☐ Page break between groups
- ☒ Summary below data

Buttons at the bottom: Remove All, OK, Cancel.

Grouping: Subtotals

Note the subtotals and expand/collapse buttons and the level buttons

The screenshot shows an Excel spreadsheet with the following data:

	A	B	C	D	E	F
	Homeroom #	First Name	Last Name	Payment Method	T-Shirt Size	
1						
2	220-A	Brigid	Ellison	Cash	Small	
3	220-B	Michael	Lazar	Cash	Small	
4	135	Anisa	Naser	Check Bounced	Small	
5	220-A	Christopher	Peyton-Gomez	Check	Small	
6	220-A	Malik	Reynolds	Cash	Small	
7	220-B	Wendy	Shaw	Cash	Small	
8	105	Melissa	White	Debit Card	Small	
9	105	Esther	Varon	Pending	Small	
10				Small Count		8
11	105	Nathan	Albee	Check	Medium	
12	220-B	Samantha	Bell	Check	Medium	
13	110	Matt	Benson	Money Order	Medium	
14	105	Christiana	Chen	Check Bounced	Medium	
15	110	Gabriel	Del Toro	Cash	Medium	
16	105	Sidney	Kelly	Check	Medium	
17	220-B	Avery	Kelly	Debit Card	Medium	
18	220-A	Chevonne	Means	Money Order	Medium	
19	135	James	Panarello	Check	Medium	
20	135	Chantal	Weller	Debit Card	Medium	
21				Medium Count		10
22	110	Kris	Ackerman	Money Order	Large	
23	105	Derek	MacDonald	Cash	Large	
24	110	Regina	Olivera	Pending	Large	
25	135	Jordan	Weller	Money Order	Large	
26	135	Alex	Yuen	Wrong Amount	Large	
27				Large Count		5
28	220-A	Juan	Flores	Pending	X-Large	
29	220-B	Tyrese	Hanlon	Debit Card	X-Large	
30	220-B	Karla	Nichols	Money Order	X-Large	
31	135	Lia	Richards	Cash	X-Large	
32				X-Large Count		4

Transposing Data

This becomes this!

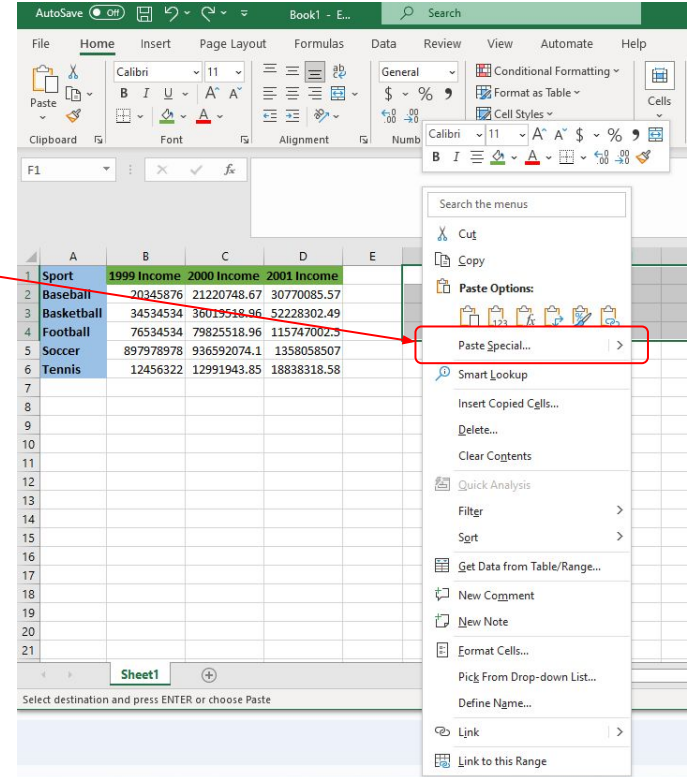
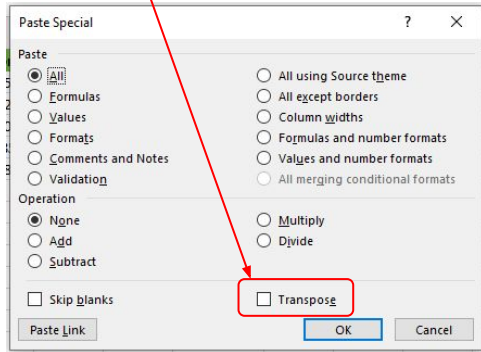
[illegible]

Transposing Data

Select data & copy

Right click upper left cell of destination and select Paste Special...

Check Transpose and click OK



String Manipulation

CONCATENATE(first_thing,
second_thing)

You'll work through this in lab!

Text Functions

- Text functions are very important when editing and manipulating data
- The following are some important text functions:

Text Function Table	
Function	Description
= Left(text,num_chars)	Returns x number of characters from a cell, starting from the left
= Right(text,num_chars)	Returns x number of characters from a cell, starting from the right
= Mid(text,start_num,num_chars)	Returns a specific number of characters from a text string starting at the position you specify
= Find(find_text,within_text,start_num)	Finds one text string within another text string and returns the number of the starting position of the found string (case sensitive)
= Len(text)	Returns the number of characters in a text string
= Upper(text)	Converts a text string to all uppercase letters
= Lower(text)	Converts a text string to all lowercase letters
= Proper(text)	Converts a text string to proper case so that the first letter in each word is uppercase and all other letters are lowercase
= Text(value,format_text)	Converts a value to text in a specific number format
= Trim(text)	Removes all spaces from a text string except for single spaces between words

<https://www.degroote.mcmaster.ca/files/2014/08/DeGroote-Data-Manip-Pres-2014-08.pdf>

Date Functions

The main function to calculate dates in Excel:

- › [DATE function](#)

Get current date and time:

- › [TODAY](#) - returns today's date
- › [NOW](#) - returns the current date and time

Convert dates to / from text:

- › [DATEVALUE](#) - converts a date in the text format to date format
- › [TEXT](#) - converts a date to a text value

Retrieve dates in Excel:

- › [DAY](#) - returns the day of the month
- › [MONTH](#) - returns the month of a specified date
- › [YEAR](#) - returns the year of a specified date
- › [EOMONTH](#) - returns the last day of the month
- › [WEEKDAY](#) - returns the day of the week
- › [WEEKNUM](#) - returns the week number of a date

Calculate date difference:

- › [DATEDIF](#) - returns the difference between two dates
- › [EDATE](#) - returns a date N months before or after the start date
- › [YEARFRAC](#) - calculates the fraction of the year between 2 dates

Calculate workdays:

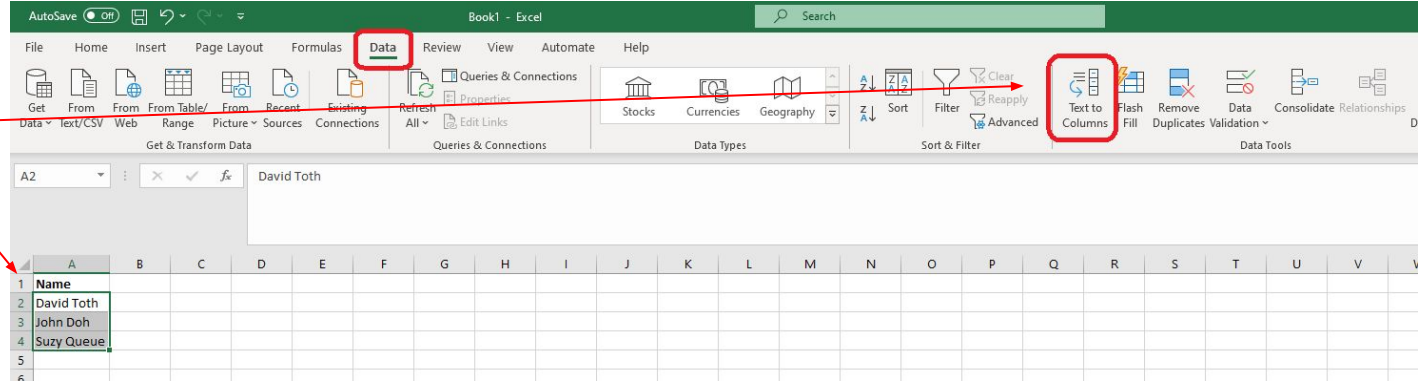
- › [WORKDAY](#) - returns a date N working days in the future or in the past
- › [WORKDAY.INTL](#) - returns a date N weekdays from the start date with custom weekends
- › [NETWORKDAYS](#) - returns the number of workdays between two dates
- › [NETWORKDAYS.INTL](#) - returns the number of workdays between two dates with custom weekends

You'll work through this in lab!

<https://www.ablebits.com/office-add-ins-blog/excel-date-functions/>

Splitting Columns

Select data to split, click Text to Columns button, fill in form



Convert Text to Columns Wizard - Step 1 of 3

The Text Wizard has determined that your data is Delimited.
If this is correct, choose Next, or choose the data type that best describes your data.

Original data type

Choose the file type that best describes your data:

☒ Delimited - Characters such as commas or tabs separate each field.
☐ Fixed width - Fields are aligned in columns with spaces between each field.

Preview of selected data:

2 David Toth
3 John Doh
4 Suzy Queue
5
6

Cancel < Back Next > Finish

Lab Time!

Lab activity is linked from Moodle.

Go back to previous labs and do UNIQUE and also the Trimming Off the Unwanted Character section on slide 60

Excel Day 4

Day 4 Topics

Pivot Tables

Pivot Tables

Why?

Provide summary of large data set

Ex: large jar of candies (thousands)

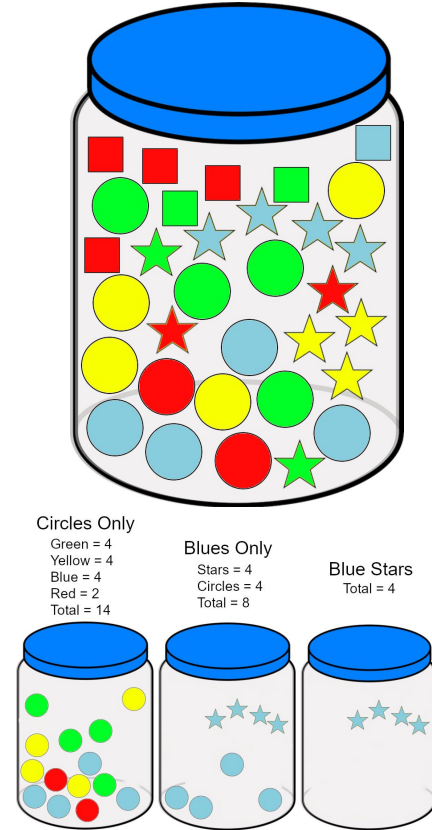
Want to know

How many red?

How many circle?

How many red square?

Pivot tables allow grouping/aggregating



Pivot Table Uses

Summarizing & rearranging large data sets into table to allow drawing conclusions

Examples

- Summarizing yearly business expenses

- Showing average spending power of different customer demographics

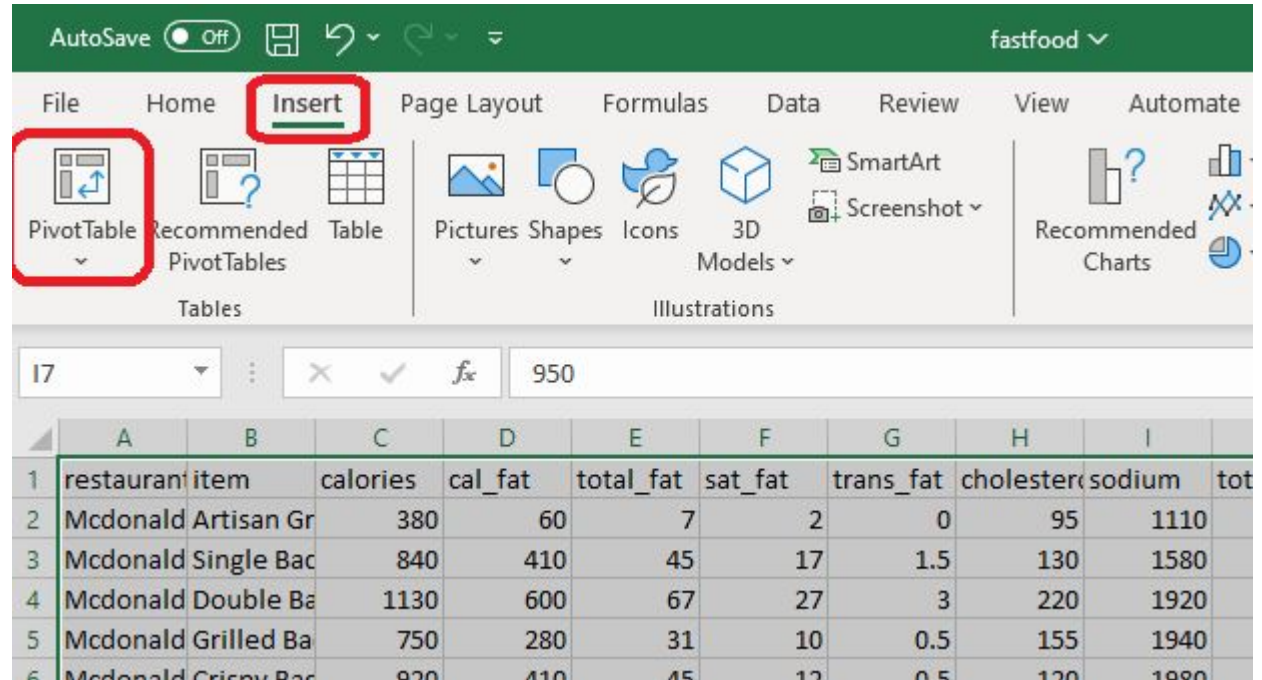
- Task count by state (not started, completed, etc)

- Count of sales by product line

Building A Pivot Table 1

Select data

Insert->Pivot Table

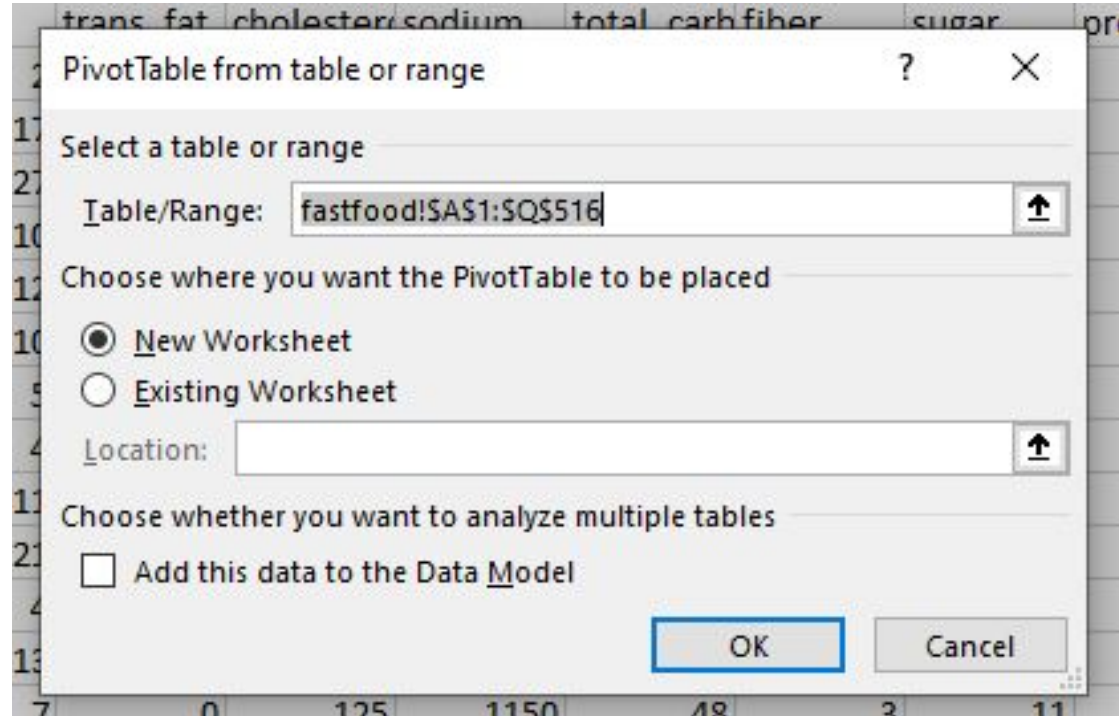


The screenshot shows the Microsoft Excel interface. The 'Insert' tab is selected on the ribbon. In the 'Tables' group, the 'PivotTable' icon is highlighted with a red box. Below the ribbon, a portion of a data table is visible, showing columns for restaurant items and their nutritional values.

	A	B	C	D	E	F	G	H	I	
1	restaurant	item	calories	cal_fat	total_fat	sat_fat	trans_fat	cholesterol	sodium	total
2	Mcdonald	Artisan Gr	380	60	7	2	0	95	1110	
3	Mcdonald	Single Bac	840	410	45	17	1.5	130	1580	
4	Mcdonald	Double Ba	1130	600	67	27	3	220	1920	
5	Mcdonald	Grilled Ba	750	280	31	10	0.5	155	1940	
6	Mcdonald	Crispy Bac	820	410	45	12	0.5	120	1880	

Building A Pivot Table 2

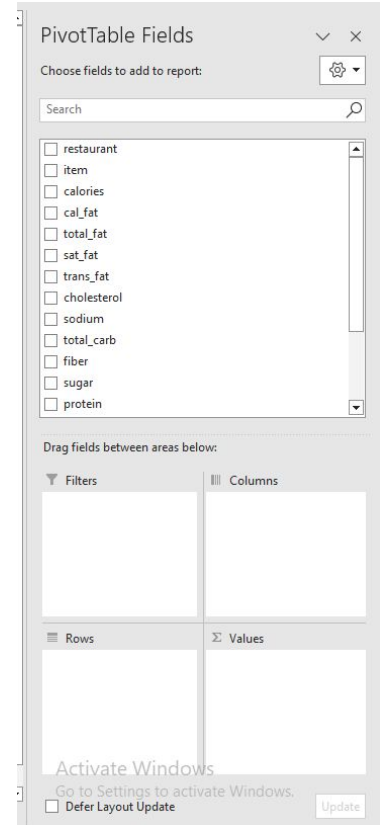
Adjust settings



Building A Pivot Table 3

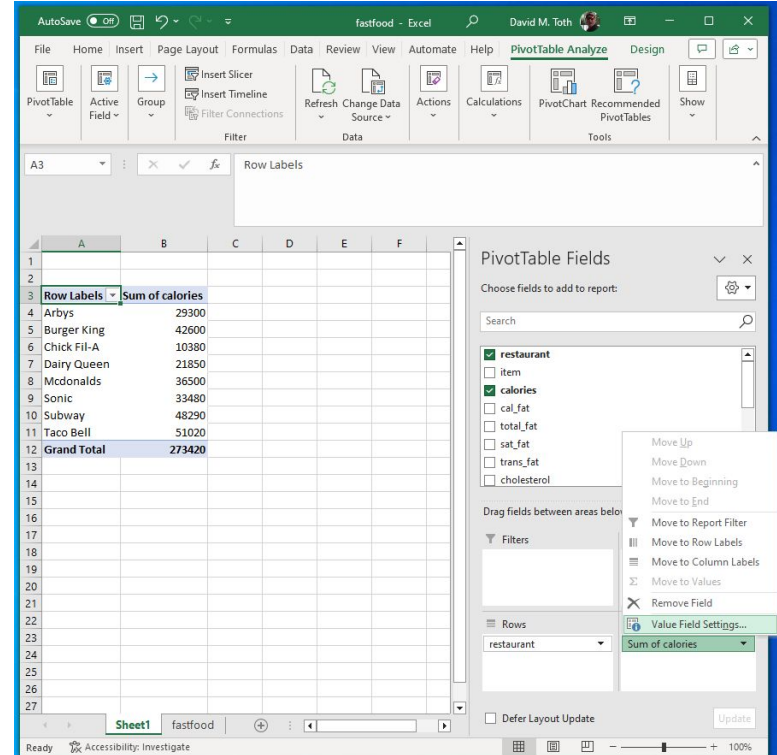
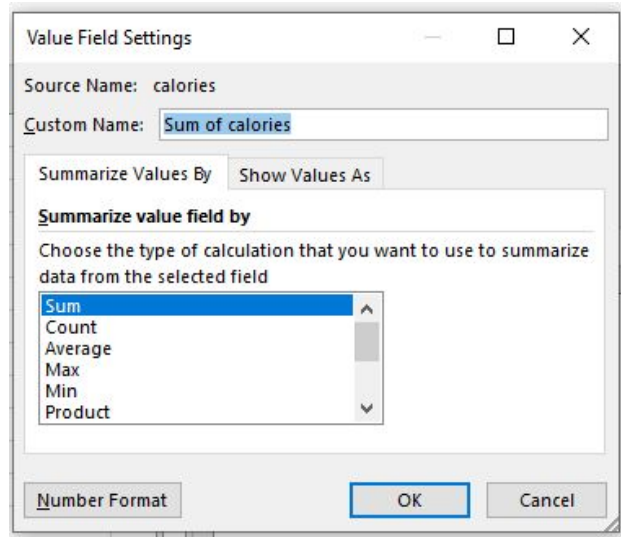
Check the variables you want or drag variables into boxes

Which box variable goes into depends on question to answer



Building A Pivot Table 4

Set calculation



Building A Pivot Table 5

Right click the table (not the header) and select a sort if desired

The screenshot shows the Microsoft Excel interface with a PivotTable. The PivotTable is located in the range A3:A12, with row labels in column A and data in columns B through F. The data includes restaurant names and their menu items. The 'PivotTable Fields' task pane is open on the right, showing the 'restaurant' field in the Filters area and the 'calories' field in the Values area. The 'Sort' option is selected in the context menu, and the 'Sort A to Z' option is chosen.

Excel Ribbon: File, Home, Insert, Page Layout, Formulas, Data, Review, View, Automate, Help, PivotTable Analyze, Design

Task Pane: PivotTable Fields

Choose fields to add to report:

- ☒ restaurant
- ☐ item
- ☒ calories
- ☐ cal_fat
- ☐ total_fat
- ☐ sat_fat
- ☐ trans_fat
- ☐ cholesterol

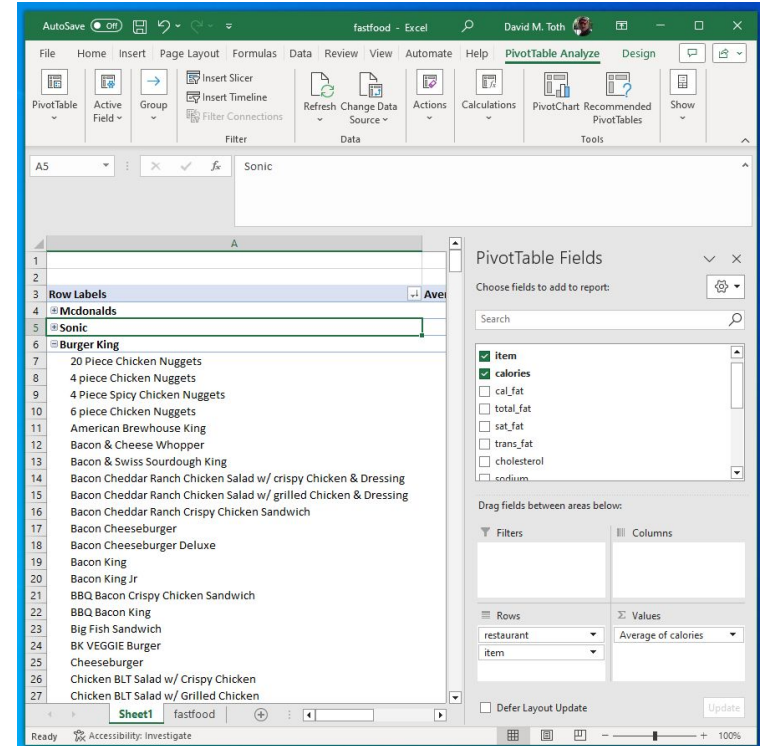
Drag fields between areas below:

Filters	Columns

Rows	Values
restaurant	Average of calories

Building A Pivot Table 6

Adding more items to rows



Building A Pivot Table 7

Adding more items to values to allow sorting by more columns

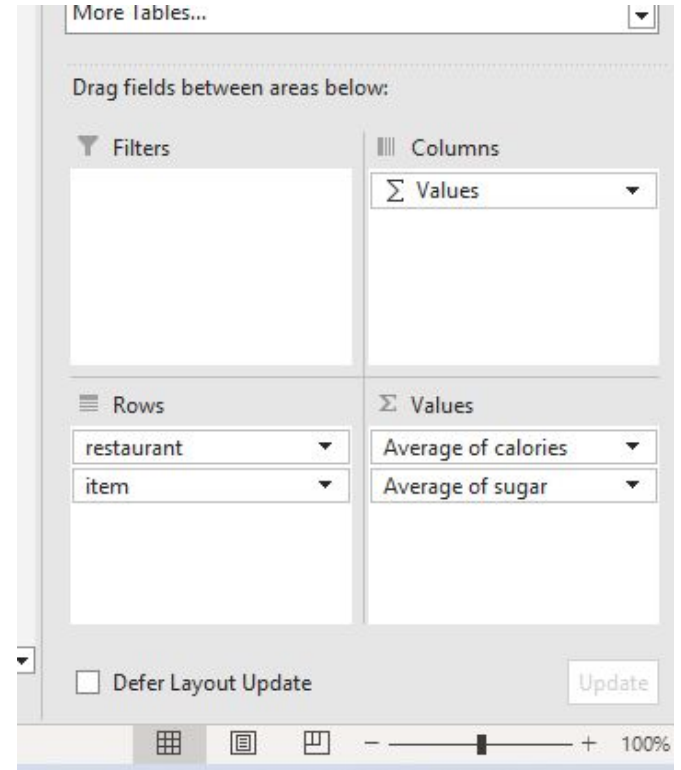
The screenshot shows an Excel spreadsheet with a PivotTable summarizing data from a table. The PivotTable is located in the range A38:G39. The data source is a table with columns A, B, and C. The PivotTable has two rows: 'McDonalds' and 'Subway'. The values are calculated as the sum of the data in columns B and C.

	A	B	C
4	McDonalds	640.3508772	11.07017544
5	Subway	503.0208333	10.09375
6	Burger King	608.5714286	8.185714286
7	Arbys	532.7272727	7.563636364
8	Sonic	631.6981132	6.528301887
9	Dairy Queen	520.2380952	6.357142857
10	Chick Fil-A	384.4444444	4.148148148
11	Chicken Salad Sandwich	430	12
12	Regular Grilled Chicken Sub Sandwich	450	10
13	Smokehouse BBQ Bacon Sandwich	500	10
14	Spicy Grilled Chicken Sub Sandwich	430	9
15	Chicken Enchiladas Meal Kit	860	8
16	Chargrilled Chicken Sandwich	310	7
17	Chicken Parmesan Meal Kit	720	7
18	Chargrilled Chicken Club Sandwich	430	7
19	Spicy Deluxe	540	6
20	Chicken Deluxe	500	6
21	Spicy Chicken Sandwich	450	5
22	Chicken Sandwich	440	5
23	Chick-n-Slider	270	4
24	4 piece Chick-n-Strips	470	4
25	3 Piece Chick-n-Strips	350	3
26	Chargrilled Chicken Cool Wrap	350	3
27	8 piece Chicken Nuggets	260	1
28	30 piece Chicken Nuggets	970	1
29	1 Piece Chick-n-Strips	120	1
30	12 piece Chicken Nuggets	390	1
31	2 Piece Chick-n-Strips	230	1
32	12 Piece Grilled Chicken Nuggets	210	1
33	6 Piece Grilled Chicken Nuggets	110	0
34	6 piece Chicken Nuggets	190	0
35	4 Piece Grilled Chicken Nuggets	70	0
36	4 piece Chicken Nuggets	130	0
37	8 piece Grilled Chicken Nuggets	140	0
38	Taco Bell	443.6521739	3.704347826

The PivotTable Fields task pane on the right shows the following configuration:

- Choose fields to add to report: restaurant, item, calories, sugar
- Rows: restaurant, item
- Columns: Values
- Values: Average of calories, Average of sugar

Building A Pivot Table 8



Lab Time!

Lab activity is to read this page and follow along.

<https://spreadsheeto.com/pivot-tables/>

There is a link to their data file near the top of the page

The screenshot shows the top of a webpage with a green header. The header contains the 'SPREADSHEETO' logo with the tagline 'LEARN EXCEL ONLINE', and navigation links for 'About', 'Consultant', and 'Courses'. A red button on the right says 'FREE EXCEL TRAINING'. The main heading is 'How to Create a Pivot Table in Excel: Step-by-Step (2023)'. The text explains that a Pivot Table is needed for summarizing large datasets. It lists benefits like sorting, calculating sums, totals, and averages. A red box highlights a call to action: 'Aa aah! Have you downloaded our free sample workbook for this guide? Get your hands on it right now to practice along with the guide below'. Below this is a 'Table of Contents' section with a link to 'What is a pivot table?'.

SPREADSHEETO
LEARN EXCEL ONLINE

About Consultant Courses

FREE EXCEL TRAINING

How to Create a Pivot Table in Excel: Step-by-Step (2023)

If you have a huge dataset that's spread across your entire sheet, and now you want to create a summary out of it – you need a Pivot Table 🙌

Pivot Tables make one of the most powerful and resourceful tools of Excel. Using them, you can create a summary out of any kind of data (no matter how voluminous it is).

You can sort your data, calculate sums, totals, and averages and even create summary tables out of it. If you are new to the concept of Pivot Tables, you'd be jaw-dropped by the end of this article.

So you're ready? Let's go.

Aa aah! Have you **downloaded our free sample workbook** for this guide? Get your hands on it right now to practice along with the guide below 🙌

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- What is a pivot table?

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Day 2

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- Grouping
- Transposing Data
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- Splitting columns

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- Pivot tables